Quantum Biomarker Algorithms for Multimodal Cancer Data
Phase 1 Final Report (August 24, 2024)

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Executive Summary

Our Q4Bio project has made significant strides in addressing the complex challenge of cancer biomarker discovery through feature selection in multimodal data, with promising implications for both quantum computing and biomedical applications. We have successfully developed a novel, quantum-classical hybrid computational approach that enhances the feasibility and effectiveness of multimodal feature selection, paving the way for useful applications of quantum computing to benefit human health and precision medicine.

Multimodal Feature Selection – We framed cancer biomarkery discovery as an innovative feature selection problem. To select informative feature sets, we designed a combinatorial optimization scheme that incorporates information-theoretic indicators to balance the expressivity and redundancy of multimodal feature sets (Section 2.4). Higher-order correlations among features potentially capture key aspects of the underlying biology. Although it is generally infeasible for classical algorithms to incorporate this information, our quantum-classical hybrid method naturally encodes third-order correlations into the optimization. Moreover, we found evidence that this inclusion significantly improves the alignment between optimization solutions and high-quality feature sets (Figure 17), opening up an exciting possibility of quantum advantage.

Potential for Quantum Advantage – The increased complexity introduced by higher-order correlations necessitated the development of more sophisticated quantum solvers. Our simulations provide compelling evidence that quantum optimization can effectively address these challenging problems (Figure 21). We conducted comprehensive comparisons between our hybrid optimization algorithm and state-of-the-art classical solvers, revealing a potential quantum advantage in this domain.

Biomedical Insights – Our quantum-inspired approach led to unexpected insights in biomedical data analysis. For instance, we discovered small, carefully selected feature sets that performed unexpectedly well in predictive tasks, such as identifying the tissue of origin of squamous cell carcinoma samples (Figure 16). Moreover, the feature discretization and preprocessing techniques we developed to reduce quantum computational resource requirements (Section 2.3) turned out to be novel regularizers that retain useful signals while dramatically reducing noise and memory requirements. Both types of results are likely to be of broader interest, beyond their quantum computation applications. Ultimately, our Phase I results strongly suggest that biological features across modalities may contain highly related information. Hence, effective feature selection may have significant benefits in the development of viable, generalizable, interpretable models.

Phase 2 and 3 Goals – (Section 5.5) Our focus in Phase 2 will be on fine-tuning the quantum optimization methods using GPU-accelerated quantum simulators, developing advanced techniques for automatic parameter tuning, and implementing better optimization routines. In anticipation of the Phase 3 demonstration on real quantum computers, we identified two promising hardware platforms for our methods: Quantinuum and D-Wave. Quantinuum trapped ion quantum computers offer the high connectivity required by our algorithm, but may not support enough qubits for evaluation of real-world problem sizes. In contrast, D-Wave quantum annealers, despite not natively supporting 3-body interactions, contain a large number of qubits, offering a viable path to interesting experiments on larger problem sizes. Additionally, to support state-of-the-art classical simulation and software development, we obtained a quote for a new Mark III high-performance computing (HPC) cluster (8 GH200 GPUs), through preferred hardware vendor NVIDIA. We also established plans for validating models in external contexts, including public data (CPTAC) and University of Chicago prospective data (Figure 3). This will allow us to rigorously evaluate feature selection methods with respect to critical downstream modeling and learning tasks in cancer research.

Interdisciplinary Convergence – Situated at the forefront of quantum computing, multimodal oncology, and systems biology, our project demonstrates the power of deeply interdisciplinary, convergent research. The algorithms from our collaboration showcase quantum computers as potential domain-specific accelerators, and highlight tight co-design with domain experts as a means to efficiently use quantum computing resources. Our work promises not only to accelerate progress in the individual disciplines but also to pave the way for synergistic development of effective diagnostic and prognostic tools in healthcare.

1 Project Overview and Summary of Activities

Our project addresses a critical gap in cancer biology: the undefined multimodal landscape of genetic, genomic, and phenotypic changes in tumors. Many cancer biomarkers are based on single data modalities and are clinically inadequate. Correlations between various biological data modes (e.g., DNA, RNA, and pathomics) can provide crucial insights into cancer subtypes and patient-specific treatment strategies [1]. However, the vast space of multimodal features and scarcity of training data present major challenges. Current classical computational approaches, including modern machine learning techniques, face limitations in algorithmic scalability [2] and clinical interpretability [3].

To overcome these challenges, we propose an innovative hybrid classical-quantum algorithm to select robust feature sets from multimodal cancer data sets, based on a polynomial constrained binary optimization (PCBO) problem formulation. Our approach, which we refer to as the PCBO-Tournament algorithm, leverages quantum resources for efficient combinatorial optimization on subproblems. We focus on feature selection, where quantum algorithms have shown promise in identifying outliers and handling non-local correlations. By framing biomarker identification as a feature selection problem and separating it from subsequent learning tasks, we aim to develop scalable, increasingly effective algorithms that can identify informative feature sets based on higher-order correlations. This approach is expected to capture complex biological relationships between DNA, RNA, and pathomic data modes more effectively than classical methods alone. Identifying relatively small sets of informative features also promotes interpretable, clinically deployable, downstream learning. By enriching the quality of the latent space for models while reducing overparameterization issues, the approach is particularly valuable in the data-scarce regime common in cancer research, where it will ultimately lead to more robust, clinically relevant insights in cancer diagnosis and treatment.

Our project is focused on the development of hybrid quantum-classical algorithms capable of efficiently analyzing multimodal cancer data to identify biomarkers with applications to cancer classification and precision oncology. Initially, we proposed two primary research thrusts:

- **Thrust 1:** Developing and testing hybrid algorithms for accurate and efficient dimensionality reduction of multimodal cancer data sets.
- **Thrust 2:** Designing and training quantum machine learning (QML) models to efficiently learn useful latent space embeddings of multimodal cancer data.

Throughout Phase 1, as we delved deeper into developing hybrid algorithms for both *feature selection* and *QML classification models* — representing research Thrusts 1 and 2, respectively — we encountered evidence suggesting that to optimize our efforts, we should focus exclusively on hybrid feature selection algorithms. First, whereas designing competitive QML models for classification progressed slowly, we successfully made significant progress developing hybrid feature selection algorithms and identifying use cases for which the algorithms yielded competitive performance, compared to standard classical approaches. We also observed that many state-of-the-art classical algorithms for solving the feature selection problem the way we have framed it are not amenable to substantial speedups via GPU acceleration, in contrast to other classical ML algorithms, which have made astounding recent progress based on GPU architectures. Hence, we hypothesize that the crossover point to quantum advantage is much closer for feature selection than for ML classification applications. That is, a quantum computer with relatively limited resources (i.e., hundreds of qubits and the capability to faithfully implement circuits with thousands of gates) is more likely to outperform classical algorithms for specific combinatorial optimization problems, rather than for classification tasks for which effective, very large neural networks can be efficiently implemented.

As a result of our preliminary assessments, and in light of the overall goal of the Q4Bio program to develop impactful quantum algorithms with a strong potential to demonstrate advantage in the final project phase, we requested a meeting with Wellcome Leap program managers on June 26 to propose and discuss a shift in focus to feature selection. The outcome of this meeting was overwhelmingly positive, and our request to alter the original statement of work was approved. We include here an updated project GANTT chart in Figure 1 that shows the shift in focus to the hybrid PCBO-Tournament algorithms we have developed for feature selection; Table 1 contains brief descriptions of all activities.

The rest of this document is organized as follows: Section 2 summarizes the results of our completed Milestone 1 and 2 activities, followed by Section 3 which briefly covers our more recent and ongoing efforts

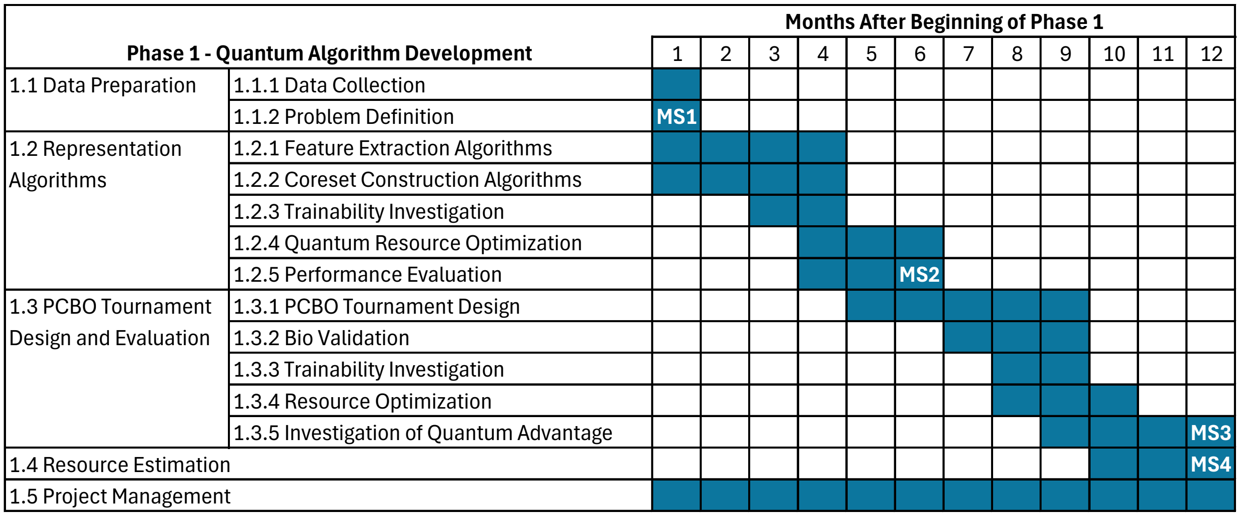


Figure 1: Project GANTT chart, updated with June 26 changes to Activities 1.3 and 1.4. “MS”: Milestone.

towards Milestones 3 and 4. (A full report regarding these milestones will be provided in the final Phase 1 project write-up.) Section 4 summarizes the main outcomes produced by our project. And finally, Sections 5 and 6 discuss how our project has met the Phase 1 evaluation criteria and the broader impacts of our work.

Sub-activity	Description	Deliverables
1.1.1 Data Collection	Gather multimodal data (DNA, mRNA, pathomics) from TCGA and ISPY.	Curated data sets (Tab. 2 and Fig. 3).
1.1.2 Problem Definition	Define relevant learning tasks: cancer tissue of origin and HPV status classification, and treatment response prediction.	Summary whitepaper (Tab. 2).
1.2.1 Feature Extraction Algorithms	Design hybrid feature extraction algorithms to select small, informative feature sets from collected data sets.	Figure reporting classification accuracy (Fig. 10).
1.2.2 Coreset Construction	Design hybrid coreset construction algorithms to reduce the sample and feature dimensionality of the data.	Figure reporting classification accuracy (Fig. 4 and 7).
1.2.3 Trainability Investigation	Assess trainability of variational quantum models through empirical evaluation and analysis of the spectrum Fisher information matrix.	Figures depicting model gradients and Fisher spectrum (Fig. 13 and 14).
1.2.4 Quantum Resource Optimization	Optimize the total quantum resources required by the hybrid algorithms through cross-layer codesign techniques.	Code describing the optimizations and a figure depicting the resources required (Fig. 11 and 12).
1.2.5 Performance Evaluation	Compare feature sets selected by our hybrid algorithms against those selected by standard classical approaches.	Report classification accuracy and computational resources (Fig. 10 and Tab. 3).
1.3.1 PCBO Tournament Design	Design scalable hybrid feature selection algorithms for multimodal cancer data.	Code describing the algorithm implementation and a figure reporting accuracy (Alg. 1, Figs. 15, 16, 17).
1.3.2 Biological Validation	Ensure that selected features reflect meaningful biology.	Figures summarizing selected features (Tab. 4).
1.3.3 Trainability Investigation	Continue the trainability evaluation of Activity 1.2.3, applied to PCBO tournament algorithms.	Comparison of various variational circuit optimizers (Fig. 22).
1.3.4 Resource Optimization	Optimize total quantum and classical resources required to implement PCBO Tournament algorithms.	Code describing the optimizations and figures displaying their effectiveness (Fig. 11, 12, and 22).
1.3.5 Investigation of Quantum Advantage	Compare the performance of the hybrid algorithms against state-of-the-art approaches and estimate the point at which a computational advantage may be achieved.	Summary whitepaper containing figures showing performance and resource requirements (Figs. 18, 19, 20, 21).
1.4.1 Resource Estimation	Develop theoretical models to capture resource requirements of the developed hybrid algorithms.	Figure displaying algorithm resource requirements at a range of problem sizes (Fig. 23).

Table 1: Phase 1 project activities and deliverables, post June 26 changes to Activities 1.3 and 1.4.

Data set name	Data Summary	Target learning problem	Relevant Milestones
`HNSC`	584 head and neck squamous cell carcinoma patients; 128 binary DNA features; 10,283 integer mRNA features; variable number of 768-dimensional pathomics feature vectors per patient	Binary HPV status classification	1,2
`Pansquamous-1`	1,427 squamous cell carcinoma patients; 18,499 binary DNA features; 13,525 integer mRNA features	3-tissue classification	1,2
`Pancan`	8,013 cancer patients; 19,533 binary DNA features; 20,530 integer mRNA features	30-tissue classification	3,4
`Pansquamous-2`	660 squamous cell carcinoma patients; 2,720 binary DNA features; 18,879 integer mRNA features; 2,046 integer pathomics features	3-tissue classification	3,4
`ISPY`	981 breast cancer patients; 10,574 integer mRNA features; 5 binary clinical features; 24 integer clinical features	Binary prediction of treatment outcome	3,4

Table 2: Overview of multimodal data sets and learning tasks used in Phase 1 algorithm development.

2 Milestones 1 and 2 Summary

Our first project milestones spanned the initial six months of Phase 1. During this time, our research encompassed three main areas: data set curation, exploration of data summarization algorithms, and the development of hybrid feature selection algorithms. In the following sections, we provide an overview of the work completed during this time and also provide some perspective on how this work has informed our ongoing research towards Milestones 3 and 4, aimed at further developing our hybrid feature selection algorithm and its applications.

2.1 Overview of Data Sets Considered

Throughout Phase 1, our project has focused on developing hybrid quantum-classical algorithms to analyze multimodal cancer data sets. Our approach has been to explore multiple diverse data sets and learning targets, to allow robust evaluations and comparisons of our hybrid algorithms against standard classical approaches across various scenarios.

As part of Sub-activities 1.1.1 and 1.1.2, we collected and curated data sets from publicly available databases, for algorithm testing and evaluation. All data sets, except `ISPY`, were derived from The Cancer Genome Atlas (TCGA)[1] online database and are summarized in Table 2. The `ISPY` data set was collected from the NCBI Gene Expression Omnibus [4], whose treatment arm structure is visualized in Figure 2. While the number of patients (rows) and features (columns) varied across data sets, they consistently incorporated at least two of these three basic data modalities:

1. Binary DNA mutations, indicating the presence or absence of single-nucleotide polymorphisms (SNPs).
2. Integer-valued mRNA expression features (values in the range 0–4), derived, using techniques we developed, from continuous expression values (Section 2.3.3).
3. Integer-valued pathomics features (values in the range 0–3), extracted by combining a published deep learning pipeline with custom methods developed by our team to enhance representation efficiency on limited qubit resources, described in Section 2.3.2.

[1] https://www.cancer.gov/tcga

Arm	Treatment	# patients
Ctr	Paclitaxel	178
	Paclitaxel + Trastuzumab	31
N	Paclitaxel + Neratinib	113
MK2206	Paclitaxel + MK-2206	60
	Paclitaxel + MK-2206 + Trastuzumab	34
Trebananib	Paclitaxel + AMG 386	115
	Paclitaxel + AMG 386 + Trastuzumab	19
VC	Paclitaxel + ABT 888 + Carboplatin	71
Pembro	Paclitaxel + Pembrolizumab	69
Ganitumab	Paclitaxel + Ganitumab	103
Ganetespib	Paclitaxel + Ganetespib	92
Pertuzumab	Paclitaxel + Pertuzumab + Trastuzumab	44
TDM1/P	T-DM1 + Pertuzumab	52

Figure 2: ISPY partitions patients into arms based on a combination of clinical features. Some of these arms are further partitioned into distinct treatment groups.

Table 2 provides a comprehensive summary of the data sets employed, including their relevance to specific project milestones. This table serves as a reference point throughout the report, illustrating the evolution of our research focus.

In addition to data used for algorithmic development, we have leveraged our unique physician-integrated team to collect additional, complementary data resources for eventual real-world performance validation, summarized in Figure 3. In particular, our team has obtained IRB clearance and identified more than 200 squamous cell carcinoma cases with associated human consent for tissue research. We are in the process of performing DNA and RNA sequencing on these samples, to mirror the modes already in the trial data. We have rigorously harmonized the sequencing pipelines, so this new data will be valuable in concert with data already undergoing analysis.

2.2 Overview of Phase 1 Project Trajectory

We initiated our study with methodologically important, but clinically relatively well-established, learning problems, such as the prediction of positive or negative status of human papillomavirus (HPV) infection, a common cause of certain cancers. In these scenarios, all tested feature selection algorithms successfully identified small feature sets (containing as few as two features) that enabled a simple logistic regression model to achieve over 90% accuracy. As our research progressed, we deliberately increased the complexity of the learning problems to challenge and refine our algorithms, while maintaining sufficiently trustworthy benchmark data. We developed algorithms that use mutual information and related concepts to optimize feature sets, as described in Section 2.4.1, which required specific feature processing. The progression in complexity involved both incorporating additional data modalities and transitioning to more complex, clinically interesting tasks, such as multi-class classification of cancer tissue of origin and, ultimately, prediction of treatment outcome. This trajectory reflects our objective of developing hybrid feature selection algorithms to address computationally challenging, clinically relevant problems.

2.3 Modality-Specific Feature Processing

To efficiently represent different modalities and compute fundamental quantities, such as mutual information and interaction information, used by our algorithms, as described in Section 2.4.1, we developed custom strategies for processing RNA and pathomics features. While the DNA and RNA data sets generally fall into a "data scarce" regime — meaning that $N_{samples} << N_{features}$ — the pathomics modality enjoys an

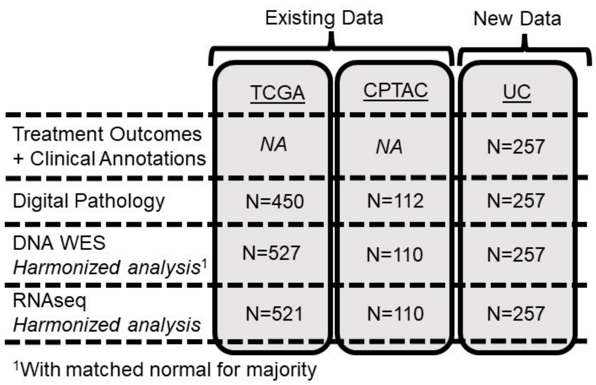


Figure 3: Summary of public data used in algorithmic development (TCGA) and planned future validation (CPTAC), and data being collected by UChicago for future real-world performance validation (UC).

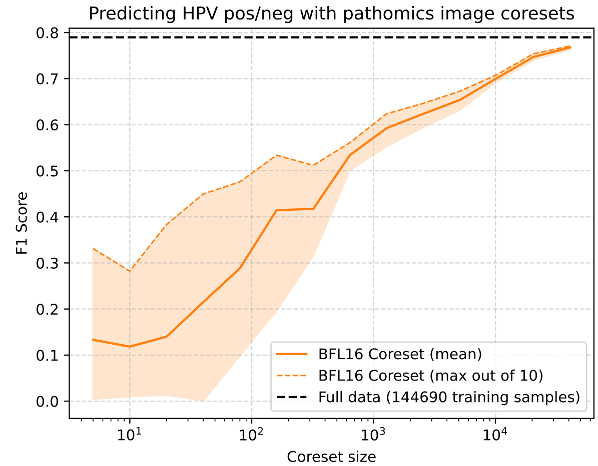


Figure 4: Constructing BFL16 coresets [5] of increasing sizes. As the size of the coreset increases, the quality of the approximation approaches that of the full training data set.

markedly greater number of features. The generation of each patient's pathomics data typically involves a pipeline of many steps, beginning with gigapixel images taken of the tumor slides. Those digitized slide images (on the order of gigabytes in size) are divided into a number (ranging from tens to thousands) of fixed-size tiles, where the number may vary across patients. In our pipeline, these tiles are passed through a self-supervised neural network, which extracts real-valued 768-dimensional feature vectors [6]. The end result is a data set containing a tensor of tens to thousands of 768-dimensional vectors for each patient. Pathomics thus stands in contrast to the two genomics modalities as "data rich", and it has the added challenge of involving a variable number of feature vectors per patient. These challenges are well-aligned with Subactivity 1.2.2, where we explored the application of coreset construction and other data summarization algorithms to enable more efficient downstream analyses.

2.3.1 Exploration of Coreset Algorithms

To investigate how to efficiently handle the large, variable data in the pathomics modality, we employ the BFL16 coreset algorithm [5]. This algorithm, previously studied for quantum-assisted clustering [7], generates an (α, β) bicriterion approximation via D^2 sampling, followed by weighted sampling to construct the coreset. We tested the performance of the coreset algorithm on the binary HNSC (head and neck squamous cell carcinoma) data set, which contains 584 patients for a total of 289,380 pathomics images, each labelled as either HPV positive (1) or negative (0). The full 289,380 pathomics images were split into 50:50 training and testing sets, both highly imbalanced; for example, the training set contains 15,621 postitive data points and 129,069 negative data points. The baseline model performance was set by a logistic regression classifier, fit to the training set, and the F1 score was obtained by evaluating that model on the test set. The results, shown in Figure 4, indicate that the data-rich pathomics modality can be effectively summarized.

2.3.2 Pathomics Feature Extraction and Dimensionality Reduction

Ultimately, to address the variable number of pathomics features per patient, we leveraged and customized several deep learning models, including a deep learning pipeline developed in prior work by our team [6]. The approach proved effective for standardizing pathomics features for use by our hybrid feature selection algorithms.

First, to generate feature vectors for each patient, we leveraged the efficient whole-slide image processing pipeline in SlideFlow [6], a state-of-the-art computer vision platform developed by our team for optimizing

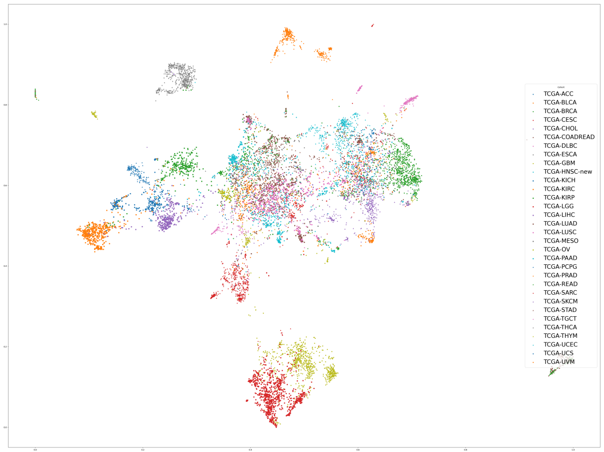


Figure 5: UMAP of continuous slide-level features in `Pancan`, the pan-cancer TCGA dataset, using Prov-GigaPath feature extractor

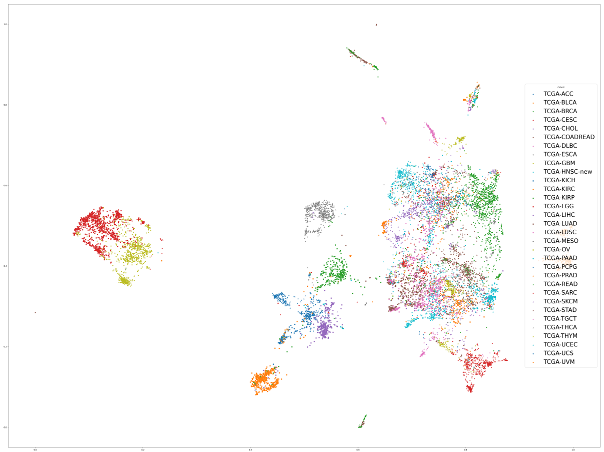


Figure 6: UMAP of discretized slide-level features in `Pancan` using Prov-GigaPath feature extractor to ensure tractability with limited qubit resources

histopathology-based biomarker development. Using Slideflow, we extracted 256x256 pixel (10x magnification) tiles for patients from all cancer subtypes in The Cancer Genome Atlas data set. These extracted tiles were then stored in TFRecord format, which allows for fast and efficient data loading during subsequent processing. To ensure efficient use of limited downstream quantum resources, we developed and tested methods, as well as assessed externally developed methods, to aggregate tiles into a single whole-slide representation. Ultimately, we identified an effective combination strategy using both Prov-GigaPath [8], a recent large-scale pathology foundation model, and our own approaches for further dimensionality reduction.

Developed by Microsoft Research and released in May 2024, Prov-GigaPath is a state-of-the-art whole-slide foundation model pretrained on a vast, diverse data set of real-world pathology images. To align with the pre-processing requirements of Prov-GigaPath while maintaining our ability to efficiently compute and customize their pipeline, we implemented support for their model pipeline within Slideflow. With the pre-processed tiles, we used Prov-GigaPath to generate tile-level embeddings that capture rich histological features learned by the model during its pretraining. To generate slide-level embeddings from these tile-level features, we implemented the aggregation method of Prov-GigaPath within Slideflow. Using a second transformer network, called LongNetViT, this method contextualizes embeddings across all tiles to create an aggregate slide-level embedding.

To reduce the bit requirements for embedding representations, we studied the distributional patterns of the features across slides within each cancer subtype and evaluated how well a variety of discretization techniques retained underlying information for downstream classification tasks. We found quartile discretization to be an effective trade-off between information retention and bit reduction.

To validate our discretization approach, we employed Uniform Manifold Approximation and Projection (UMAP) to visualize both the continuous and discrete representations of the pan-cancer dataset `Pancan`. The resulting projections showed remarkably similar patterns, indicating that the discretized features retained the essential structure of the data Figure 5 and Figure 6. We further validated this by comparing classifier performance using logistic regression on both the full-size continuous feature sets and the discretized features to predict tissue of origin (among lung, cervix, and hand and neck) in the pan-squamous data. The results demonstrated nearly identical performance across all metrics. For the continuous features, we achieved an accuracy of 0.88, with macro-averaged precision, recall, and F1-scores of 0.86, 0.84, and 0.84 respectively. The discretized features, suitable for quantum processing, yielded the same 0.88 accuracy, with macro-averaged precision, recall, and F1-scores of 0.86, 0.84, and 0.85 respectively. Both approaches also achieved a weighted average F1-score of 0.88 across all classes. These results strongly support the efficacy of our discretization method in preserving the predictive power of the original continuous features while enabling efficient quantum processing.

The final output of our pipeline for each patient is one representative pathomic feature vector of size 1x768,

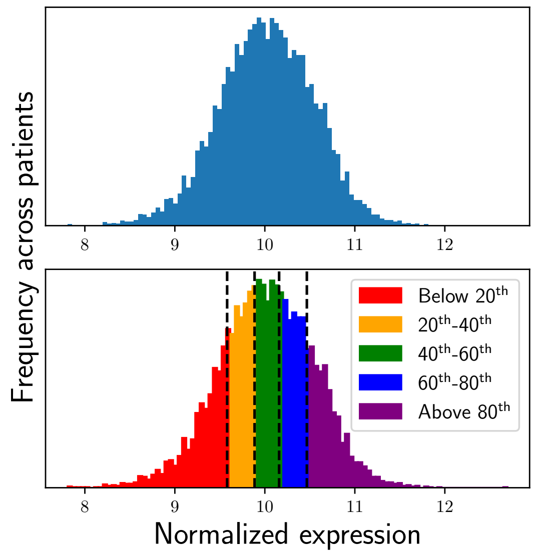


Figure 7: Example of the mRNA discretization for the DHX8 gene feature in the `Pancan` data. The continuous expression values (top) were each assigned one of five categorical values (bottom) according to percentile bins defined by multiples of 20 percent.

containing integer values in the range 0–3, which indicate low, medium, high, and very high values associated with each feature axis of the tissue image, ultimately enabling normalized, condensed, and informative custom representations of pathomics data for downstream tasks.

2.3.3 Discretization of mRNA expression data

Although, unlike pathomics, mRNA is data scarce with respect to patient sample numbers in these data sets, some level of data summarization and discretization enables more efficient, stable computation of quantities, such as mutual information, that are key for our feature selection algorithms. To this end, we transformed mRNA expression values into discrete variables. For each gene, the expression values across all patients were partitioned into five bins, defined by the 20^{th}, 40^{th}, 60^{th}, and 80^{th} quantiles, resulting in integer values in the range 0–4 for each gene; an example for the gene DHX8 in the `Pancan` data is shown in Figure 7. For `Pansquamous-1`, `Pancan`, and `Pansquamous-2`, the expression data that were discretized are the $\log_2$ transformation of the `scaled_estimate` values from Illumina HiSeq[2] available in the GDAC portal. For the `ISPY` data, the expression data that were discretized are the ComBat-combined microarray data available on the NCBI Gene Expression Omnibus[3]. The discretization was performed on these `ISPY` microarray data after removing all genes which did not have gene expression value greater than 9 in at least five patients, in order to account for nonspecific binding of transcripts to the microarray [9]. `Pansquamous-1`, `Pancan`, and `Pansquamous-2` data were discretized without any pre-processing other than that described in the TCGA portal.

While this preprocessing technique was designed for the purpose of more easily computing mutual information between features, our recent manuscript shows that it also enables the selection of small, informative feature sets in a multimodal subset of the `Pansquamous-1` data that predict tissue of origin remarkably well, even better than state-of-the-art feature selection algorithms [10]. This finding is highly nontrivial; the size and high-dimensionality of high-throughput data collection techniques, coupled with the complexity of genetic and transcriptomic interactions and their effect on tissues, makes the selection of small, informative feature sets for tasks like tissue of origin and cancer prognosis notoriously difficult [11]. Therefore, the contribution of this preprocessing technique is not merely a computational convenience, but also a means of

[2] https://gdac.broadinstitute.org/runs/stddata__2016_01_28/samples1_report/index.html
[3] https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE194040

ensuring better selection of informative feature sets using feature selection algorithms on both classical and quantum hardware.

2.4 Development of the PCBO-Tournament Algorithm

Our primary focus throughout Phase 1 has been the development of hybrid algorithms for the analysis of multimodal cancer data sets. After identifying feature selection as a promising application domain for potential quantum advantage, we developed the novel hybrid algorithm that we refer to as the PCBO-Tournament algorithm. In the following subsections, we summarize the key aspects and project activities pertaining to the development of this algorithm.

2.4.1 Framing Feature Selection as a PCBO Problem

As a part of Sub-activity 1.2.1 we identified feature selection in multimodal cancer data as a promising area, with the potential to demonstrate quantum advantage in the near future. The structure of the feature selection problem is a natural match for the types of problems where quantum computers are likely to realize computational advantages. To illustrate this point, consider a multimodal cancer data set, which typically contains relatively few patients (rows) but many features (columns), together with an objective of solving some learning problem (e.g., tissue of origin classification) with high accuracy. The goal of feature selection is to select a subset of N features out of all F features that can be used to solve the learning problem, while retaining high accuracy. Solutions to his problem have a succinct representation — as binary strings of length F and Hamming weight N — but there are $\binom{F}{N}$ possible solutions — i.e., an exponentially growing search space. In fact, prior work has proposed various methods for applying quantum computers to feature selection [12, 13, 14].

Mutual information in feature selection. Our approach to feature selection is partly motivated from a biological perspective, as prior work has demonstrated the effectiveness of mutual information (MI) for measuring feature relevance and reducing redundancy among features in biomedical data sets [15, 16]. MI as a filter technique can identify a broad set of potentially informative features, which can then be refined using more selective methods. Combining MI with other feature selection approaches has been shown to maximize the advantages of individual techniques while mitigating their limitations, leading to more robust and biologically relevant feature sets for tasks, such as cancer biomarker identification [11] [17].

Problem formulation. The feature selection task, over F total features, can be expressed mathematically as a PCBO problem of the form:

$$\begin{aligned}&\arg\min_{\boldsymbol{x}} \left(\sum_{i}^{F} C(i)x_i + \sum_{i<j}^{F} C(i,j)x_i x_j + \sum_{i<j<k}^{F} C(i,j,k)x_i x_j x_k + \dots \right) \\ &\text{subject to the constraints: } |\boldsymbol{x}|_1 = N \in \mathbb{Z}, \quad \boldsymbol{x} \in \mathbb{Z}_2^N\end{aligned} \tag{1}$$

Here, the aim is to "turn on" N bits of the binary vector $\boldsymbol{x}$ which correspond to the selected feature set. The functions $C(\cdot)$ yield the coefficients defining the optimization problem. We use the mutual information between features and class labels, as well as between pairs of features, to define the coefficient values for the feature selection problem. As a concrete example, Equation 2 shows a specific PCBO definition, with real-valued hyperparameter λ, which we refer to as minimum-redundancy maximum-relevancy (`mRmR`):

$$Q_{\texttt{mRmR}}(\boldsymbol{x}, \lambda) = -\lambda \sum_{i} I(f_i; y)x_i + \sum_{i,j} I(f_i; f_j)x_i x_j \tag{2}$$

Recall that the mutual information between two random variables A and B is defined as [18]:

$$I(A;B) = \sum_{a\in A, b\in B} P_{(A,B)}(a,b) \log \frac{P_{(A,B)}(a,b)}{P_A(a)P_B(b)}. \tag{3}$$

Algorithm 1 PCBO-Tournament

Input: Data set $D = \{f_0, \ldots, f_{F-1}, y\}$, Num qubits n, final feature set size N
Pre-processing:
$\text{PCBO}_{coefs} \leftarrow$ `compute_pcbo_coefs`(D)
Hybrid Divide-and-Conquer Tournament:
$T \leftarrow \{f_0, \ldots, f_{F-1}\}$
while $|T| > n$ **do**
  $S \leftarrow$ `generate_subproblems`(T, PCBO_{coefs})
  $T_{temp} \leftarrow \{\}$
  for s_i in S **do**
    $\text{selected}_{features} \leftarrow$ `solve_pcbo`(s_i)
    append $\text{selected}_{features}$ **to** T_{temp}
  end for
  $T \leftarrow T_{temp}$
end while
$s \leftarrow$ `generate_final_pcbo`(T, PCBO_{coefs})
$\text{final}_{features} \leftarrow$ `solve_pcbo`(s)
assert $|\text{final}_{features}| = N$
Return $\text{final}_{features}$

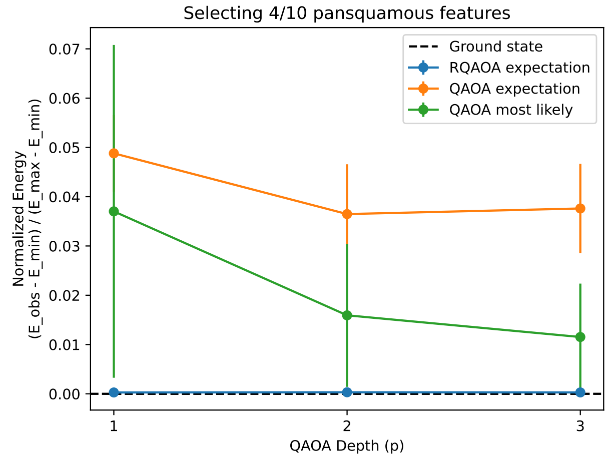


Figure 8: RQAOA is a non-local solver which leverages multiple rounds of QAOA optimization to obtain high quality solutions to the feature selection optimization problems.

And since the objective is to minimize $Q(\boldsymbol{x}, \lambda)$, Equation 2 can be understood intuitively to reward the selection of features f_i that are highly correlated with the labels y of the learning problem, and to punish the selection of pairs of highly correlated features.

As indicated by the PCBO definition in Equation 1, the problem may generally include higher-order terms corresponding to interactions between arbitrary numbers of features — this leads to a delicate trade-off between the computational cost of solving the problem and the biological accuracy of the problem definition. On one hand, higher-order feature correlations in feature selection are biologically relevant, for example, due to complex genetic regulatory networks [15, 16]. On the other hand, solving PCBOs containing these higher-order interactions significantly increases computational complexity [19, 20]. Therefore, our approach leverages a non-local quantum optimization algorithm to obtain higher-quality solutions to the higher-order PCBO problems that better match the underlying biology, resulting in the selection of more robust and informative feature sets.

2.4.2 Hybrid Solvers for PCBO Problems

The specific quantum algorithm we use to find low energy solutions to the feature selection PCBO problems is called the Recursive Quantum Approximate Optimization Algorithm (RQAOA) which was introduced in prior work from IBM [21]. This algorithm recursively evaluates a QAOA [22] solver on smaller subproblems in each iteration. As part of Sub-activities 1.2.1 and 1.2.5 we studied the ability of different hybrid solvers to find low energy (i.e., "good") solutions to the PCBO problems. Figure 8 shows a comparison between RQAOA and the non-recursive version of the algorithm, QAOA, for solving a 10-qubit PCBO problem. The y-axis shows the energy E_{obs} of the solution obtained by the RQAOA or QAOA algorithm normalized with respect to the maximum and minimum energies of the worst and best solutions, respectively. Please find additional details included in our Milestone 2 Deliverables. Through evaluations like that shown in Figure 8, we find that the recursive aspect of RQAOA is necessary to adequately solve the PCBO problems, and our ongoing Milestone 3 and 4 evaluations are aimed at scaling up these simulations to study larger, more interesting problems.

2.4.3 Specification of the PCBO-Tournament Algorithm

We have now covered each of the basic building blocks of the full PCBO-Tournament algorithm, save one — a divide-and-conquer step which enables a quantum device with a fixed number of qubits to select features over data sets with an arbitrary number of features. We discuss the algorithmic tradeoffs introduced by this divide-and-conquer step in further detail in Section 2.6.1. Here, we briefly summarize the key steps of the PCBO-Tournament algorithm:

1. Classical pre-processing of multimodal cancer data: Utilizing the summarization techniques discussed in Section 2.3, we prepare the data for quantum processing. This step also involves precomputing the PCBO coefficients $C(\cdot)$ that appear in Equation 1.

2. Classical divide-and-conquer: To enable quantum solvers to handle large-scale feature selection problems, we implement classical strategies to decompose the problem into manageable subproblems. This step involves breaking up the original F features into subsets of size less than or equal to the number of qubits available on the quantum hardware.

3. Recursive quantum optimization: We employ RQAOA to solve the PCBO defined by each subproblem. This quantum component allows us to explore high-dimensional search spaces and capture higher-order feature interactions more effectively than classical methods.

A pseudocode outline of the full algorithm is provided in Algorithm 1. Algorithm 1 appears in our unpublished manuscript "On the potential for quantum advantage in feature selection applications for multimodal cancer data", which contains many further details about each of the steps in the PCBO-Tournament algorithm [10]. We plan to release this manuscript on the `arXiv` and submit it for peer review towards the end of Phase 1; at present, we have made the manuscript available for confidential use via our shared Google Drive with Wellcome Leap.

2.5 Performance Evaluation of the PCBO-Tournament Algorithm

As a part of Sub-activity 1.2.5, we evaluated the performance of the PCBO-Tournament algorithm through two main avenues. First, we sought to establish that the low-energy solutions obtained by solving the PCBOs with RQAOA correspond to high-accuracy feature sets for a given learning problem. This analysis was only feasible for relatively small problems (containing 10s of features) where brute force search for the optimal solution and simulation of the RQAOA quantum circuits are still feasible. Then, we evaluated the PCBO-Tournament algorithm by selecting feature sets from the `Pansquamous-1` data set listed in Table 2. For these evaluations, where each PCBO subproblem contained approximately 1000 features, we use simulated annealing as an efficient classical stand-in for the RQAOA approach. Our ongoing Milestone 3 and 4 evaluations, as part of Sub-activity 1.3.5, include performance comparisons between RQAOA and simulated annealing within the simulatable regime and indicate that the classical algorithms struggle to find great solutions to PCBO problems that contain at least third-order terms. Therefore, we may assume that the substitution of simulated annealing for RQAOA in the larger scale evaluations is a rough approximation or lower bound on overall performance.

2.5.1 Correlating PCBO Solution Energy to Feature Set Accuracy

In our feature selection manuscript [10], we studied the hypothesis that low energy PCBO solutions correspond to high accuracy feature sets. We have included the results of that evaluation here in Figure 9. To briefly summarize our methodology: we first generated "seeded" feature sets composed of three "good" features — which were identified by a random forest-based feature importance algorithm applied to the full `Pancancer` data set — and 10 randomly sampled features (ensuring no overlap with the good features). Then, we constructed PCBO problems over these 13 features (with the desired number of selected features set to three) for different settings of the λ hyperparameter (see Equation 2, this hyperparameter controls the relative weight of the feature-label correlations). A priori, we would expect a good feature selection algorithm to find the seeded features in this reduced data set since they were specifically chosen to classify the data with higher accuracy than the randomly chosen features. We evaluated this in a brute-force manner by generating all bit strings of length 13 and Hamming weight three — corresponding to all possible feature sets of size three on this reduced data set. For each bit string, we compute its energy value given by the PCBO specification, and the classification accuracy achieved by a logistic regression model trained on the corresponding feature set. Figure 9 includes results for three different PCBO specifications including `mRmR`

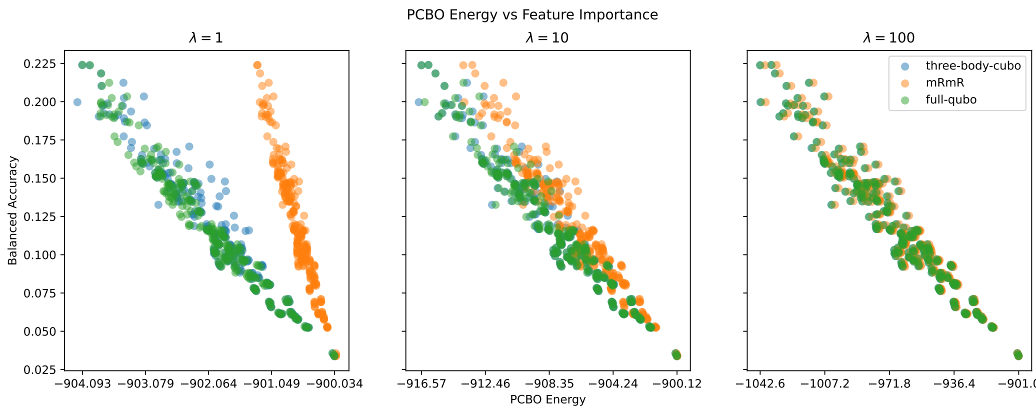


Figure 9: Results of brute-force evaluation of all sets of three features on a seeded subset of 13 features drawn from the full `Pancancer` data set. For every possible 13-choose-3 feature set, we compute the resulting classification accuracy as well as its PCBO energy. We consider three different PCBO definitions with varying values of λ. The hyper parameter value λ determines the emphasis on correlation with the label, which is why the definitions appear to become equivalent as λ increases. The high negative correlation between the low PCBO energies and the high accuracy values indicates the utility of mutual information-based filter methods for feature selection.

Table 3: Algorithm Scaling Comparison for data sets with M rows, N columns. Q denotes the number of qubits, and C is the complexity of the chosen clustering algorithm.

Algorithm	Type	Time Complexity
`feature importance` [23]	Classical	$\mathcal{O}(MN \log M)$
`univariate` [24]	Classical	$\mathcal{O}(MN)$
`mifs-nd` [25]	Classical	$\mathcal{O}(N^2)$
`PCBO-Tournament`	Hybrid	$\mathcal{O}(\frac{N}{Q}C)$

(whose definition is given in Equation 2) as well as `full-qubo` and `three-body-cubo` – for brevity, we do not include the explicit definitions of these PCBOs here but they can be found in our manuscript [10].

The results of this evaluation clearly show the relationship between the bit string (i.e., feature set) PCBO energies and their classification accuracy on the `Pancancer` 30-tissue classification task. There is a clear correlation between a lower PCBO energy and a higher feature importance score. This correlation indicates that finding low energy solutions to PCBO problems can produce high quality feature sets without needing explicit knowledge of the downstream classification score. Furthermore, such a correlation implies that algorithms which find better PCBO solutions will also produce more accurate feature sets.

2.5.2 Evaluating PCBO-Tournaments on Realistic Multimodal Data Sets

We briefly summarize our comparative performance analysis of the hybrid PCBO-Tournament algorithm against classical methods for predicting cancer tissue of origin in the `Pansquamous-1` data set. These results were obtained as part of Sub-activity 1.2.5, and appeared in our manuscript [10]; our ongoing Milestone 3 and 4 activities are applying a similar analysis to the other data sets in Table 2.

We evaluated the performance of our feature selection algorithm on the `Pansquamous-1` data set where the learning task is to predict the cancer tissue of origin – among cervical (CESC), head and neck (HNSC), and lung (LUSC) squamous cell carcinoma – for each patient. Figure 10 contains the results of this evaluation where we plot the weighted F1 score (the geometric mean of precision and recall weighted by the relative sizes of each class) for feature sets ranging in size from 4 to 200 features. This graph demonstrates that our hybrid quantum-classical approach (solid lines) consistently outperforms traditional methods (dotted lines) in selecting small, informative sets of features for cancer classification. We compare against classical fea-

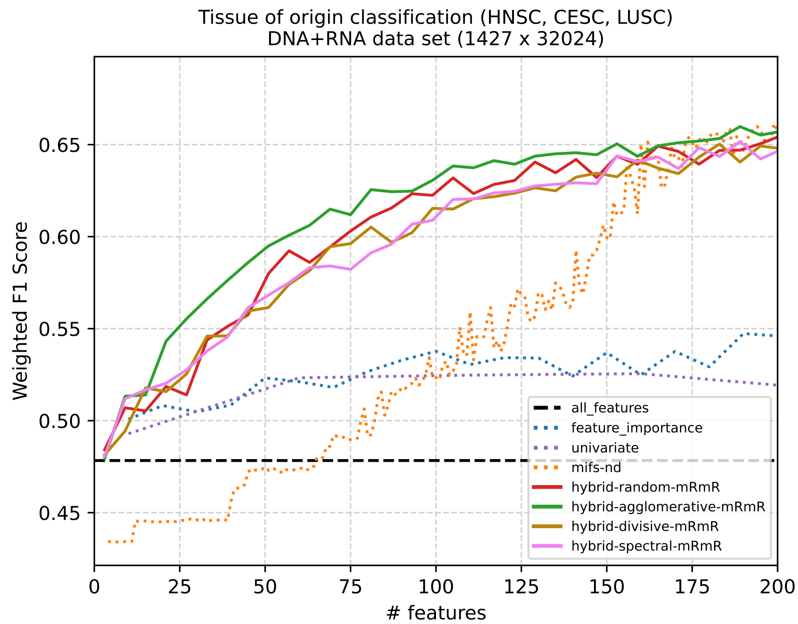


Figure 10: Empirical evaluation of classical (dashed lines) and our hybrid (solid lines) feature selection algorithms. All algorithms were applied to a variant of the combined `Pansquamous-1` data set containing 32,024 total features. In this variant, the 13,525 mRNA variables were transformed and discretized so as to function as distractor variables. Each algorithm produces a feature set of a specific size (X-axis) which is input to a fixed logistic regression classifier, and we perform 5-fold cross-validation on the data set to produce the reported F1 Score (Y-axis, higher is better).

ture selection algorithms (dotted lines) including: feature importance (based on the `ExtraTreesClassifier` from `scikit-learn` with default parameters [24]), a univariate approach based on feature-target mutual information (using the `SelectKBest(mutual_info_classif)` function from `scikit-learn` with default parameters), and `mifs-nd` which is a greedy algorithm based on feature-feature and feature-target mutual information [25]. These are compared against four separate implementations of the PCBO Tournament (solid lines): one for random subproblem generation and one for each of agglomerative, divisive, and spectral clustering. These four implementations are alike in that the PCBO coefficients are computed according to the `mRmR-qubo` approach (Equation 2).

For each point N along the X-axis, every feature selection algorithm is asked to produce a feature set of size N. Each of these feature sets is used to train a one-versus-rest logistic regression classifier (implemented using `scikit-learn` with default parameter settings) and the F1-score of the classifier is obtained using 5-fold cross validation. The black, horizontal, dashed line shows the performance of the same classifier applied to the full 32,024 feature data set. The relatively low performance seen in this case highlights the value of feature selection techniques for reducing noise and boosting useful information contained in a training data set.

In the small feature set regime ($N \leq 150$) the PCBO-Tournament algorithm outperforms all of the classical feature selection algorithms. Beyond this size, the mutual information-based `mifs-nd` algorithm produces feature sets that match the performance of the hybrid algorithms. These results demonstrate the potential utility of the PCBO-Tournament to select smaller, more accurate feature sets than purely classical methods. We are continuing to study and evaluate the performance of this approach, testing other PCBO definitions and evaluating on additional data sets as part of Milestone 3 and 4, and these results will be included in the final Phase 1 report.

Table 3 contains the asymptotic resource complexity of the feature selection algorithms considered here. The resource complexities are given with respect to a data set containing M total samples and N features per sample. The PCBO-Tournament algorithm also depends on the number of qubits Q available as well

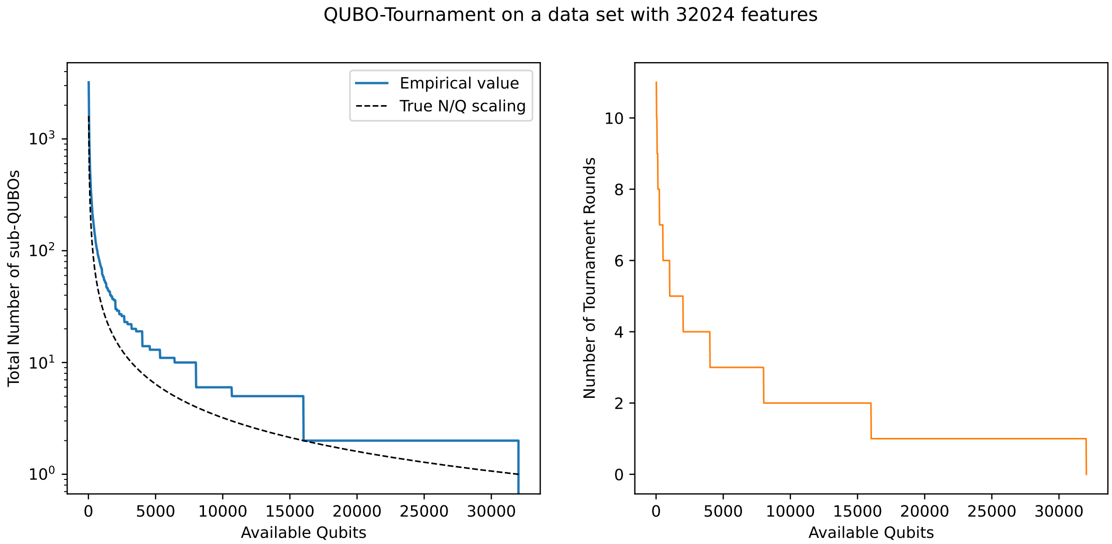


Figure 11: The PCBO-Tournament algorithm takes a divide-and-conquer approach to feature selection which allows for a quantum device, with a limited number of qubits, to select features on a data set of arbitrary size (in these plots we set $N = 32024$). This induces a tradeoff between the number of qubits available for solving the PCBO problems and the number of tournament iterations and total subproblems required to process a full data set. The left hand side shows the number of PCBO subproblems appearing in each tournament round, and the right side shows the total number of rounds required.

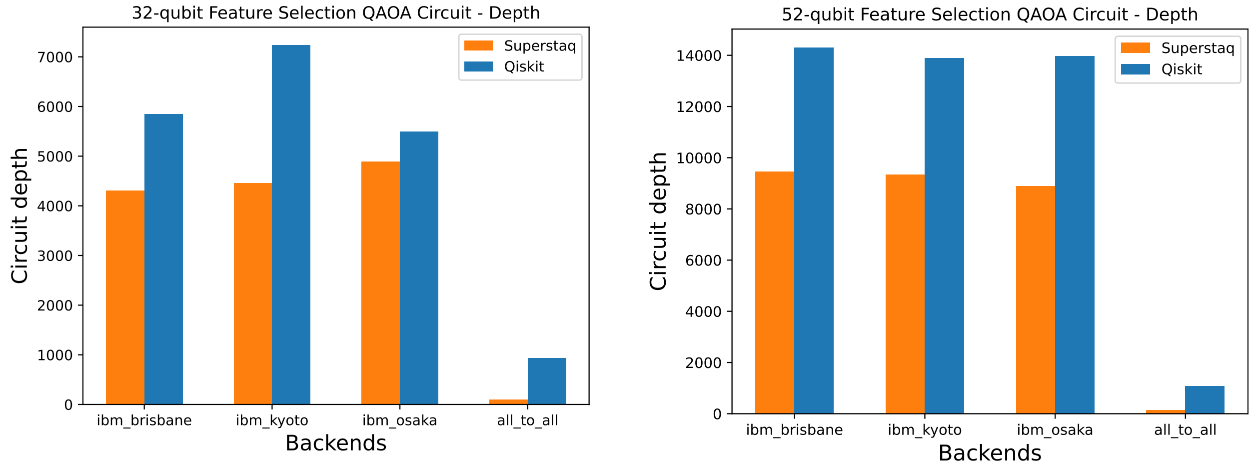
as the complexity C of the classical clustering algorithm that is chosen. In Figure 10 we consider random groupings of features as well as agglomerative, divisive, and spectral clustering methods based on the mutual information distance matrices produced between feature pairs.

2.6 Cross-layer Resource Optimization

Sub-activities 1.2.4 and 1.3.4 are both aimed at optimizing the total quantum and classical resources required to implement the PCBO-Tournament algorithm. This is an area where our team has extensive expertise and much of our prior work has focused on the codesign of quantum computing algorithms and applications [26, 27, 28]. In this section we focus on two primary examples of cross-layer optimization — one at the upper layers of the stack and the other at the lower layers. Later, in Sections 3 and 5, we briefly mention ongoing and future work in this vein of quantum codesign applied specifically to the PCBO-Tournament algorithm.

2.6.1 Algorithmic-level tradeoffs

Optimistically, even in the near- to far-future qubits will continue to be a scarce and valuable resource relative to the classical bits available on classical co-processors. The divide-and-conquer aspect of the PCBO-Tournament algorithm was introduced specifically to handle the limited number of qubits available on today's and future quantum hardware. This is an algorithmic-level tradeoff that enables a quantum device with a fixed number Q of qubits to select features from a data set containing a total of N features by only considering a maximum of Q features at any single time. The tradeoff is that this means multiple subproblems, scaling like N/Q, must be solved in order to produce the final feature set. In general, this is a favorable tradeoff since, as the plots in Figure 11 show, only a handful of tournament rounds are needed even when the qubit count is very small. The alternative being that the user must wait until an N qubit processor is available to evaluate an N feature data set. Finally, we note that the PCBO subproblems appearing in each tournament round may be trivially parallelized if multiple Q-qubit processors are available.



(a) (b)

Figure 12: Compilation using the Superstaq and Qiskit frameworks to decompose a feature selection QAOA circuit for a PCBO containing single- and second-order terms. A variety of target backends are shown with varying connectivity, and across each the Superstaq compiler achieves shorter depths.

2.6.2 Hardware-aware circuit compilation

In Section 2.4.3 we saw that the PCBO-Tournament algorithm is composed of iterative rounds of: classical clustering of features, followed by quantum optimization. In this section we focus on hardware-aware compiler optimizations that may be applied to the quantum circuits generated as a part of the quantum optimization step. These evaluations were performed as part of Sub-activity 1.2.4 and additional details and plots can be found in our Month 6 (Milestone 2) deliverables.

We consider both the Superstaq [26] and the Qiskit (with `optimization_level=3`) quantum circuit compilers tasked with mapping example QAOA circuits to hardware backends with varying connectivity. In Figure 12 we report the circuit depths of 32- and 52-qubit QAOA circuits after compilation via Superstaq or Qiskit. We include results both for nearest-neighbor connected superconducting devices as well as all-to-all connected ion-trap devices. This highlights (a) the extra overhead incurred when mapping the feature selection QAOA circuits to a device with restricted qubit connectivity, and (b) the versatility of the Superstaq framework which can target a variety of backends including superconducting, trapped ion, and neutral atom platforms [26]. The Superstaq compiler is able to consistently produce shorter depth circuits due to its pulse-level optimizations that exploit the native gate set supported by the target hardware backend to better decompose the circuit gate operations.

2.7 Trainability of Variational QAOA Circuits

As part of Sub-activity 1.2.3 we investigated the trainability of the QAOA circuits produced within the PCBO-Tournament algorithm. We evaluated the trainability on the basis of empirical measurements of the gradient during optimization as well as an analysis of the eigenvalue spectrum of the Fisher information matrix defined by the QAOA circuits. Additional details and plots can be found in our Month 4 deliverable document.

Figure 13 shows an example from this trainability evaluation, focusing on a specific 10-qubit case. The plot shows the expected energy (left) and gradient magnitude (right) throughout the training process for 100 randomly initialized trials where many of the trials often become stuck in sub-optimal local minima. Additionally, we visualize the eigenvalue spectrum of the Fisher information matrix defined by the QAOA circuits as a histogram shown in Figure 14. This is a useful means of investigating the trainability of quantum models since the eigenvalues of the Fisher information matrix are related to the curvature of the optimization landscape [29]. The histograms in Figure 14 were computed using open source code [30] and indicate that the optimization landscape is flat along many dimensions (indicated by eigenvalues with magnitudes close

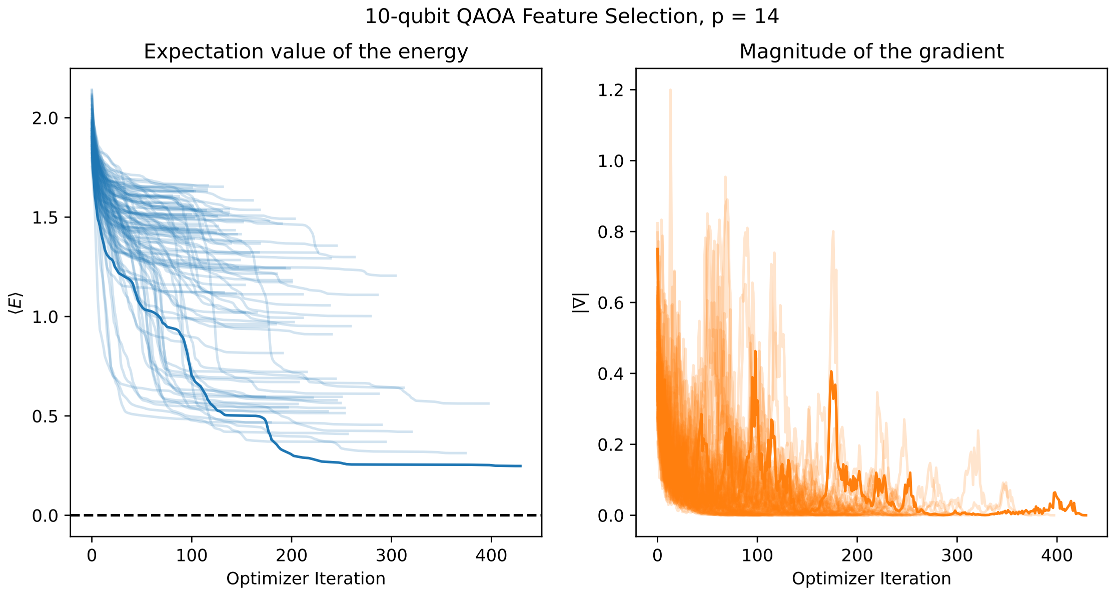


Figure 13: The value of the loss function and gradient magnitude measured throughout the training, using the COBYLA optimizer, of 100 randomly initialized QAOA circuits with $p = 14$. The run which achieved the lowest energy has its loss and gradient highlighted in the darker color. The dashed black line indicates the global minimum. Additional details and trainability plots can be found in our Month 4 deliverables.

to zero) with some curvature along other dimensions (indicated by the eigenvalues with larger magnitude), and that the landscape looks progressively flatter as we increase the depth and number of parameters in the circuit (increasing the p parameter).

These results highlight the difficulty of perfectly optimizing the QAOA circuits, which is expected given the prior work demonstrating the presence of barren plateaus and local minima in the loss landscape of these variational circuits [31]. We note that our use of RQAOA does alleviate this issue to some extent — in Figure 13, reaching the ground state energy (black dashed line in left plot) would correspond to perfectly solving the PCBO problem in a single run. In general this is difficult to achieve, because of barren plateaus and local minima, but the RQAOA relaxes this onerous trainability requirement because during each round of QAOA optimization *only a single variable is fixed* [21], in this way the solutions produced by RQAOA tend to find the global optimum more often – this can also be seen in Figure 8.

Ensuring efficient and accurate optimization of the variational circuits is one of the primary challenges that must be overcome to ensure robust quantum advantage. We have identified multiple paths towards addressing this issue, including the use of global versus local optimizers, and those evaluations are a primary focus of our ongoing and planned work for Phase 2. Interestingly, very recent related work has emerged indicating that many of the trainability issues associated with the QAOA may be addressed using intelligent classical pre-computation, leveraging parameter concentration and transferability phenomena, to effectively warm start the optimization [32]. These research directions point towards the viability of advantageous applications of the QAOA using early fault-tolerant quantum computers.

3 Milestones 3 and 4 Summary

Building upon the results and deliverables from Milestones 1 and 2, our research has progressed significantly through Milestones 3 and 4. We continue to evaluate the PCBO-Tournament algorithm's efficacy in identifying small, informative feature sets across the `Pancan`, `Pansquamous-2`, and `ISPY` data sets outlined in Table 2. Our focus has been on leveraging the PCBO problem formulation to exploit multiple data modalities, particularly by incorporating higher-order correlations between feature variables. Concurrently, we are testing the RQAOA's performance on small, simulatable problem instances.

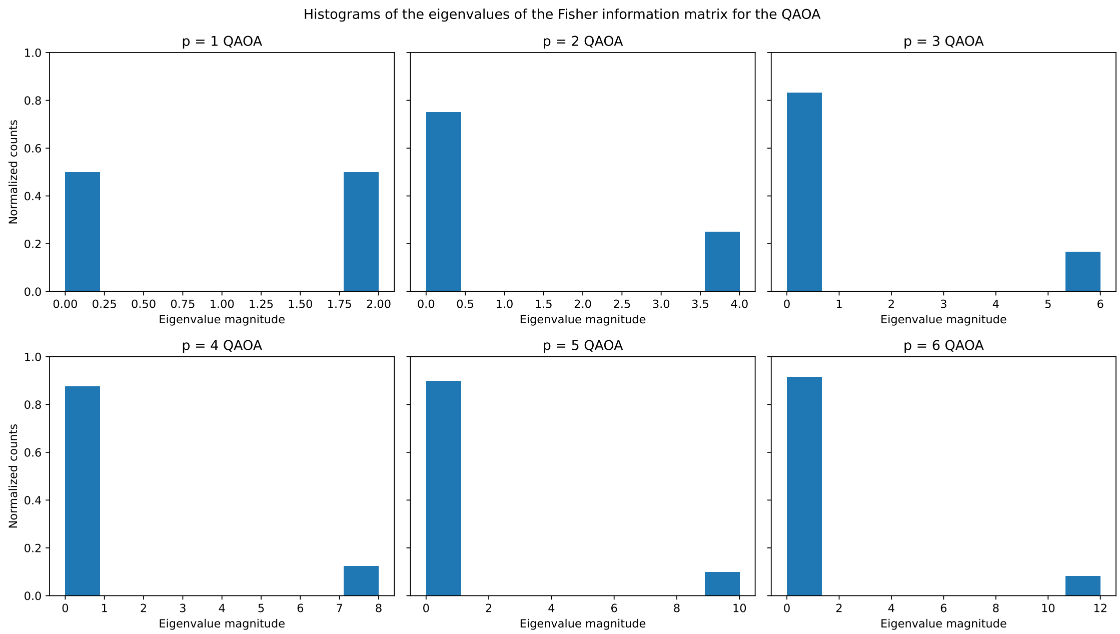


Figure 14: Distribution of Fisher information matrix eigenvalues for QAOA circuits.

3.1 Evaluating PCBO-Tournaments on the `Pansquamous-2` Multimodal Data Set

Our initial evaluation on the `Pansquamous-1` data set, in Figure 10, considered only DNA and mRNA modalities. Furthermore, the mRNA features were pre-processed to function as distractor variables. Here, we return to this data set, but, importantly, we incorporate the pre-processing techniques described in Sections 2.3.2 and 2.3.3 to produce informative pathomics and mRNA features, respectively. We integrated all of the available DNA, mRNA, and pathomics features together into a data set we refer to as `Pansquamous-2` (details in Table 2).

The `Pansquamous-2` data set represents head and neck (HNSC), cervical (CESC) and lung (LUSC) squamous cell carcinomas, a biomedical and computational setting in which our team has substantial expertise, which is also a challenging benchmark for feature selection algorithms. Comprising 23,645 features, this is a high-dimensional data set containing many complex multimodal correlations. Additionally, a logistic regression classifier trained on the full feature set achieves very high performance, with a mean accuracy exceeding 97%, establishing a formidable baseline against which to measure the performance of different feature selection algorithms.

Our experimental set-up to test the performance of the PCBO-Tournament and other comparable classical feature selection algorithms is described as follows. First, we addressed a known and problematic batch effect in the pathomic data that stems from site-specific signatures introduced during data collection [33]. This batch effect was ameliorated by stratifying the full data set into three separate folds, where the patient data from each site appear in exactly one of the three folds. These three disjoint folds were then combined to produce three different *site-stratified* train-test splits of the data. Concretely, we start with disjoint subsets $\{\texttt{fold}_1, \texttt{fold}_2, \texttt{fold}_3\}$ of the full data such that for each site, all patients from the site are contained within the same fold. Then we combine these folds to produce train-test splits, such as $\texttt{split}_1 := (\{\texttt{fold}_2, \texttt{fold}_3\}_{\text{train}}, \{\texttt{fold}_1\}_{\text{test}})$ with the other two splits generated by permuting the folds. Thus, if the learning model focuses on site-specific features in the training data, its performance suffers on the test data, where patient data are sourced from completely different sites. The stratification was also optimized to preserve, to the extent possible, the balance of cancer tissue labels (HNSC, CESC, and LUSC) seen in the full data set.

For each train-test split, we evaluated each feature selection algorithm by asking it to select a small set of features from the training data. (In results shown, target sizes of feature sets range from 6 to 14.) Then, a one-versus-rest logistic regression classifier (similar to Section 2.5.2, implemented using `scikit-learn` with

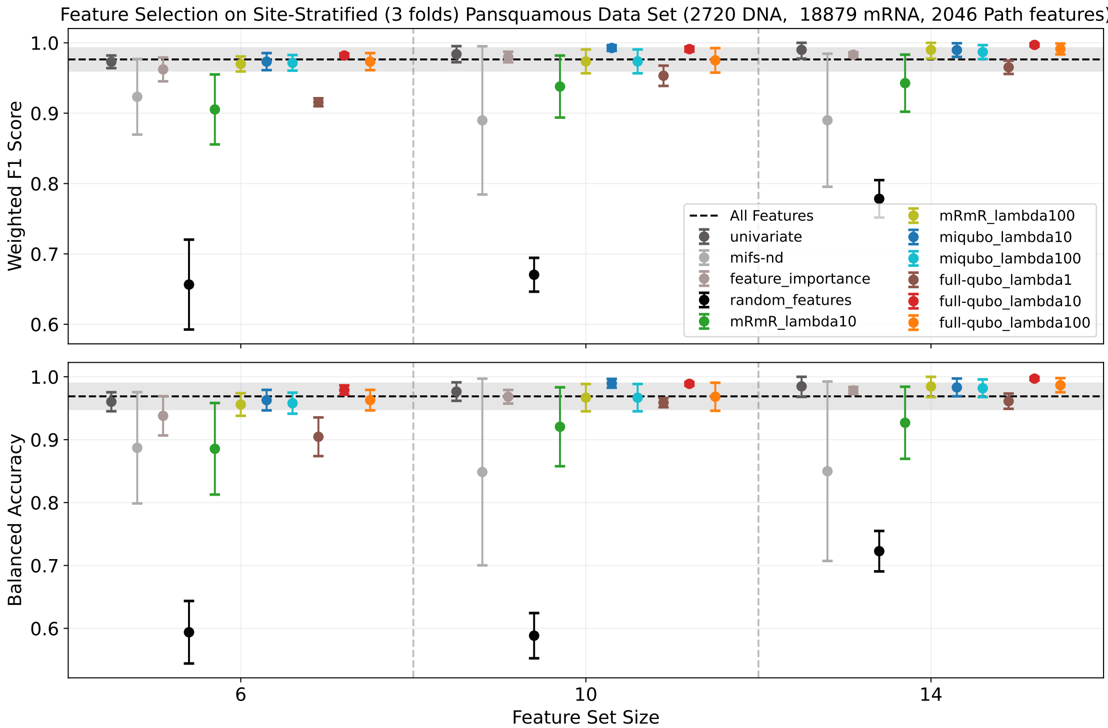


Figure 15: **PCBO-Tournament algorithms select feature sets of size 6–14 that compete with or improve on the high performance of a classifier trained on all 23,645 features.** Plots show the average performance (weighted F1 score, balanced accuracy) of one-versus-rest logistic regression classifiers trained on feature sets selected by the methods indicated by color, i.e., the PCBO-Tournament, using the `mRmR`, `miqubo`, and `full-qubo` algorithms, and other classical feature selection algorithms. Features are selected from the `Pansquamous-2` multimodal data set, and the learning target is predicting the cancer tissue of origin (one of HNSC, CESC, and LUSC). Points represent the mean performance score of the classifier across three train-test splits. Performance measures account for unbalanced class sizes. Error bars and gray horizontal band denote the 95% confidence interval of the mean. Note: Figure 16 is a close-up of the upper portion of the y-axis of this plot.

default parameter settings) was trained to predict tissue label on the training data using the selected features. The trained model was then used to classify the test data. The feature selection, training, and testing was repeated for each algorithm five times, and the results corresponding to the best balanced accuracy on the test data were stored to represent that algorithm's performance for that specific train-test split. (Balanced accuracy is used to account for large differences in class sizes.) The performance across the three train-test splits was averaged to produce the data shown in Figures 15 and 16 (which show the same data, with Figure 16 zoomed in on the higher-performing algorithms).

For comparison with PCBO-Tournament methods, we included classical feature importance [23], univariate [24], and `mifs-nd` [25] algorithms for feature selection, introduced earlier in Section 2.5.2. We tested three different PCBO definitions. In addition to `mRmR`, whose definition is given in Equation 2, we evaluated two other variants: `miqubo` and `full-qubo`.

The `miqubo` formulation [12] weights feature sets according to their correlation with the label and their conditional mutual information, as follows:

$$Q_{\texttt{miqubo}}(\boldsymbol{x}, \lambda) = -\lambda \sum_i I(f_i; y) x_i - \sum_{i,j} I(f_i; y|f_j) x_i x_j \tag{4}$$

It is important that our feature selection optimization problem is constrained to enforce a desired number of features in the final feature set. Without this constraint, the `miqubo` definition has a trivial global optimum

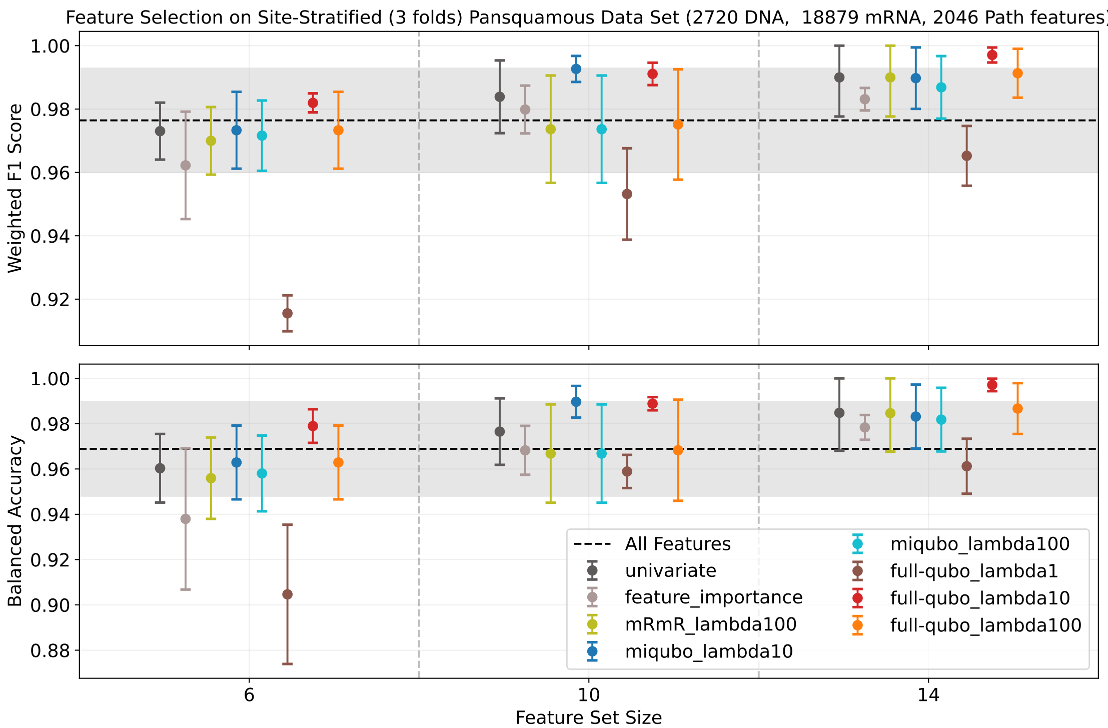


Figure 16: **The `full-qubo` PCBO-Tournament with $\lambda = 10$ selects very high-performing feature sets of size 14 that significantly improve on the all-feature classifier accuracy.** Data and annotations are as in Figure 15. Zooming in on the best performers here shows that even for small feature set sizes, the PCBO-Tournament approach performs well.

where all features are included in the feature set.

We also formulated a new PCBO that we refer to as `full-qubo`, defined as:

$$Q_{\texttt{full-qubo}}(\boldsymbol{x}, \lambda) = -\lambda \sum_i I(f_i; y) x_i + \sum_{i,j} (I(f_i; f_j) - I(f_i; y|f_j)) x_i x_j \tag{5}$$

We evalute these PCBOs using various settings of the λ hyperparameter, which controls the relative weight of the feature-label correlation term. Our results suggest that different PCBO variants require different settings of λ to achieve optimal performance. Accurately and efficiently setting hyperparameters is further discussed in Section 3.2. We also plan to more fully address this challenge in Phase 2 by leveraging larger HPC resources to search through vast swaths of parameter space.

For our evaluations, we set the maximum size of the PCBO sub-problems to 1000 variables. Since we cannot simulate 1000-qubit quantum circuits, to evaluate PCBOs at this size, we employ the classical QBSolv solver, developed by D-Wave [34].

Figures 15 and 16 show the results of our feature selection algorithm evaluations for feature sets of size 6, 10, and 14, out of a total of 23,645 features. (Figure 16 excludes underperforming algorithms to zoom in on the upper end of the y-axis.) The results show that the `full-qubo` PCBO with $\lambda = 10$ is able to select very small, predictive feature sets, which outperform those selected by other algorithms (Figure 16). With 14 features, it significantly improves the balanced accuracy beyond that of the classifier trained on all features. By successfully identifying a handful of highly informative features amidst the full feature space and leveraging them to enhance classifier performance, our approach not only simplifies the model but also yields tangible improvements in performance. This outcome underscores the capacity of PCBO-Tournament to discern subtle yet crucial patterns in complex, high-dimensional multimodal data, offering valuable insights into feature set importance and model optimization.

As part of Sub-activity 1.3.2, we examined the specific features selected by the algorithms on one of the

Algorithm	Univariate	`mifs-nd`	Feature Importance
Features	mRNA-NACA2	mRNA-TBX5	mRNA-TBX5
	mRNA-SFTPA2	DNA-TP53|7157	DNA-TP53|7157
	mRNA-PABPC3	DNA-TTN|7273	mRNA-SFTA3
	mRNA-SFTPB	PATH-277	mRNA-SFTPB
	mRNA-ATP5EP2	PATH-1222	mRNA-BMP5
	mRNA-LOC407835	PATH-1441	mRNA-GCSH
	mRNA-TBX5	PATH-1650	mRNA-POU6F2
	mRNA-C14orf19	PATH-1702	mRNA-AGTR2
	mRNA-RPL19P12	PATH-1874	mRNA-TBX4
	DNA-TP53|7157	PATH-1956	mRNA-C14orf19

Algorithm	`mRmR` $\lambda = 100$	`miqubo` $\lambda = 10$	`full-qubo` $\lambda = 10$
Features	mRNA-TBX5	mRNA-SFTPA2	mRNA-TMEM108
	mRNA-SFTPA1	mRNA-NACA2	mRNA-HIRIP3
	mRNA-ATP5EP2	mRNA-SFTPB	mRNA-PABPC3
	mRNA-LOC407835	mRNA-PABPC3	mRNA-SCARA5
	mRNA-SFTPA2	mRNA-LOC407835	mRNA-ATP5EP2
	mRNA-C14orf19	DNA-TP53|7157	mRNA-CCDC72
	DNA-TP53|7157	mRNA-TBX5	DNA-TP53|7157
	mRNA-SFTPB	mRNA-C14orf19	mRNA-LOC407835
	mRNA-PABPC3	mRNA-SFTPA1	mRNA-SFTPA2
	mRNA-NACA2	mRNA-GLUD2	mRNA-TBX5

Table 4: Example showing the 10-feature sets selected by the algorithms for one of the train-test splits used to generate Figures 15 and 16. For ease of identification, we prepend the TCGA feature identifiers with "mRNA-" and "DNA-" labels. The numbers which appear in the "PATH-" features refer to the index in the extracted pathomics feature vector described in Section 2.3.2.

train-test splits in the `Pansquamous-2` evaluation. Table 4 contains the 10-feature sets selected by a subset of the classical and PCBO-Tournament algorithms for one of the train-test splits.

The features selected by the algorithms reflect meaningful biology in several key ways. The consistent selection of TP53, a critical tumor suppressor gene mutated in approximately 50% of human cancers, underscores the algorithms' ability to identify fundamental drivers of cancer development. The inclusion of tissue-specific markers like SFTPA1, SFTPA2, and SFTPB (surfactant proteins characteristic of lung tissue) demonstrates the capture of important tissue-of-origin information. The selection of developmental genes (TBX5, TBX4) and regulators of mRNA processing (PABPC3) points to the algorithms' recognition of processes often dysregulated in cancer. Notably, the identification of TMEM108, a transmembrane protein, highlights potentially targetable molecules on the cell surface, while the selection of HIRIP3, a histone-interacting protein, indicates recognition of epigenetic factors that could be therapeutically relevant. Furthermore, the integration of pathomics features (PATH-*) alongside genetic markers shows that the algorithms can successfully combine multiple data modalities – genetic, transcriptomic, and morphological – to create a comprehensive biological profile of cancer. Broadly, the results reflect the complex nature of cancer biology and suggest that the selected features provide a biologically meaningful, relevant, and potentially clinically actionable representation of cancer processes.

3.2 Third-order PCBO energy landscapes are a better proxy for feature quality than QUBO energy landscapes in `ISPY` data

While predicting tissue of origin is handled reasonably well by classical methods (some highly parameterized) in contexts, such as pan-squamous cell carcinomas, predicting response to treatment remains a critical, immensely challenging problem in many contexts. To explore our feature selection approach in this setting, we worked with the `ISPY` dataset from a recent, adaptive clinical trial for breast cancer patients (Table 2,

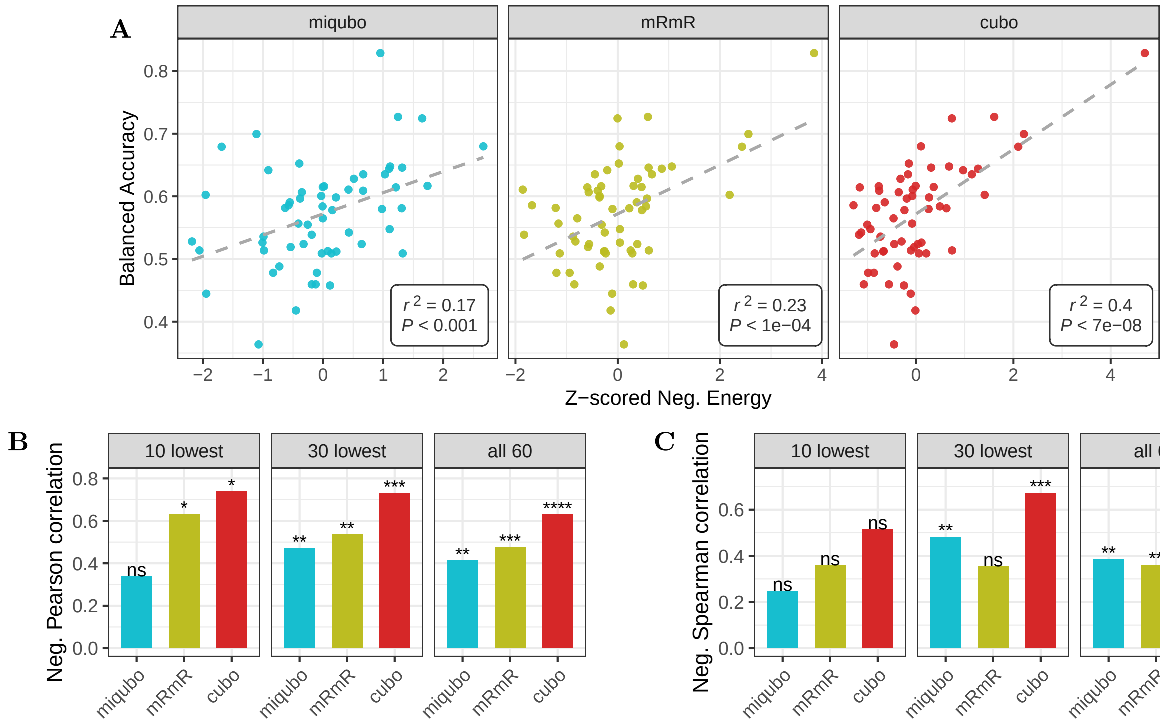


Figure 17: **The energy landscape of `tccubo` better predicts balanced accuracy of feature sets than that of `miqubo` and `mRmR`.** (**A**) Scatter plots of 60 triplets of features, plotted by the z-scored negative energy (x axis) of a PCBO (panel title), versus the balanced accuracy (y axis) of a classifier using the triplet to predicting response to cancer treatment (Sec. 3.2). Dashed line shows a linear regression fit, annotated with r^2 and p-value for the regression coefficient. (**B, C**) Bar plots show the Pearson correlation (B) and Spearman correlation (C) between the energies of each PCBO and balanced accuracy, on the full set of 60 triplets and when restricted to the triplets with the 10 and 30 lowest energies. Significance is indicated as: **** $P < 1 \times 10^{-6}$, *** $P < 1 \times 10^{-3}$, ** $P < 0.01$, * $P < 0.05$, ns: $P \geq 0.05$.

Figure 2). We investigated whether incorporating more complex biological relationships into the quantum model leads to better feature selection, potentially by uncovering important biology that simpler models might miss. The PCBOs `miqubo`, `mRmR`, and `full-qubo` can be used to learn informative subsets of features by leveraging both mutual information between feature and target, which we refer to as "first-order", and mutual information between features, which we refer to as "second-order". While we leverage these PCBOs to select small, informative feature subsets in the `Pansquamous-1` data (Section 2.5.2), second-order information is insufficient to capture potentially important types of covariation between three or more random variables, such as synergistic interactions between genes [15]. Because the PCBO-Tournament algorithm uses an arbitrary PCBO for solving subproblems (Algorithm 1), and PCBOs are well-defined for higher-order interactions, we explored PCBOs with third- and other higher-order interactions.

The second-order terms from `miqubo`, `mRmR`, and `full-qubo` can be generalized to third-order terms using *third-order interaction information*, which for random variables f_i, f_j, f_k, is defined as:

$$I(f_i; f_j; f_k) = I(f_i; f_j) - I(f_i; f_j \mid f_k),$$

Third-order information between f_i, f_j, f_k, conditioned on random variable f_l, is defined as:

$$I(f_i; f_j; f_k \mid f_l) = \mathbb{E}_{f_l}\left[I(f_i; f_j; f_k) \mid f_l\right].$$

To leverage third-order information in feature selection, we introduced a family of PCBOs containing all first-, second-, and third-order interaction information coefficients that are either unconditioned or conditioned only upon the target, which we refer to as target-conditioned cubic unconstrained binary optimization (`tccubo`) PCBOs. These take the form

$$Q_{\texttt{tccubo}}(\boldsymbol{x}, \lambda, \beta, \gamma, \delta, \epsilon) = -\lambda \sum_i I\left(f_i; y\right) x_i + \sum_{i,j} \left(\beta I\left(f_i; f_j\right) + \gamma I\left(f_i; f_j \mid y\right)\right) x_i x_j + \sum_{i,j,k} \left(\delta I\left(f_i; f_j; f_k\right) + \epsilon I\left(f_i; f_j; f_k \mid y\right)\right) x_i x_j x_k. \tag{6}$$

Using a subset of the `ISPY` data, we performed experiments to investigate the performance of `tccubo` PCBOs in selecting features sets for predicting pathologic complete response (pCR) to a specific treatment. To focus on the effect of including or excluding third-order information, we assessed the performance of algorithms in selecting a feature set of size 3, where two features in the set are fixed in advance. The results suggest that the energy functions of PCBOs with non-zero third-order information-theoretic coefficients serve as a better proxy for the balanced accuracy attainable by a classifier (Figure 17).

In more detail, the algorithm were applied using the data from the 69 patients of the Pembro arm (Fig. 2), the only one which had relatively high pathologic complete response (pCR) rates across cancer subtypes [4, Fig. 6C]. From 10,603 discrete `ISPY` features (Table 2), 1000 features were selected by applying the PCBO-Tournament algorithm, with mRmR as the PCBO, $\lambda = 10$ as the mRmR hyperparameter, and random subproblem generation (Alg. 1). For each of the $\binom{1000}{2}$ pairs of those features, we computed the balanced accuracy (using threefold cross-validation) of using only the two features to predicting pCR. The pair of features that yielded the 75^{th} percentile in balanced accuracy (which consisted of the genes *SLC16A2* and *TCF15*) was selected. Then the 1000 features were sorted in descending order by feature importance, computed using the `feature_importances_` attribute of `scikit-learn` random forest classifiers [24]. A set of 60 features was identified by selecting features with ranks 1-20, 491-510 and 981-1000 in this sorted list. Combining each of these 60 features with the pair (*TCF15* and *SLC16A2*) produced 60 triplets of features.

For each of the 60 triplets (f_1, f_2, f_3) and binary treatment outcome f_{pCR}, the coefficients

$$a := \sum_{l=1}^{3} I\left(f_l; f_{\text{pCR}}\right), \qquad b := \sum_{l=1}^{3}\sum_{m=1}^{3} I\left(f_l; f_m\right), \qquad c := \sum_{l=1}^{3}\sum_{m=1}^{3} I\left(f_l; f_m | f_{\text{pCR}}\right),$$

$$d := \sum_{l=1}^{3}\sum_{m=1}^{3}\sum_{n=1}^{3} I\left(f_l; f_m; f_n\right), \qquad e := \sum_{l=1}^{3}\sum_{m=1}^{3}\sum_{n=1}^{3} I\left(f_l; f_m; f_n \mid f_{\text{pCR}}\right)$$

were computed. Using these coefficients, the hyperparameter λ for `miqubo` (Eq. 4), the hyperparameter λ for `mRmR` (Eq. 2), and the hyperparameters $\lambda, \beta, \gamma, \delta, \epsilon$ for `tccubo` were computed using linear regression, where the dependent variable was the negative balanced accuracy of the optimal support vector machine (SVM) with radial basis function kernel[4] under threefold cross-validation.

The results show that relative to the QUBOs, low energy of tccubo better predicts high balanced accuracy for a triplet of features (Fig. 17A). The Pearson correlation of negative `tccubo` energy and balanced accuracy, both among all triplets and specifically among low-energy triplets, is significant and higher than that of either `miqubo` or `mRmR` (Fig. 17B). Since the correlation of negative energy and balanced accuracy is maximized by the hyperparameter fitting process for each PCBO, these data demonstrate a real-world setting for which the energy landscape of `tccubo` is a better proxy for balanced accuracy than `miqubo` or `mRmR` energy landscapes.

The results suggest that classical information-theoretic feature selection algorithms are constrained by their limited use of interaction information of at most the second order, and that quantum-enabled, higher-order information-theoretic feature selection tools may have an advantage in computational oncology. Further, even when restricting to the set of n triplets with the smallest `tccubo` energy for small n, the `tccubo` energy landscape still preserves monotonicity between PCBO energy and balanced accuracy better than both `miqubo` and `mRmR`, as measured by both Pearson (Fig. 17B) and Spearman (Fig. 17C) correlations. In practice, stronger monotonicity between PCBO energy and balanced accuracy makes it more likely that a PCBO solver will select the feature set with the lowest balanced accuracy.

[4] While logistic regression is used in other experiments, SVM with radial vasis function kernel is used here due to having strictly more expressivity as a classifier than logistic regression classifiers.

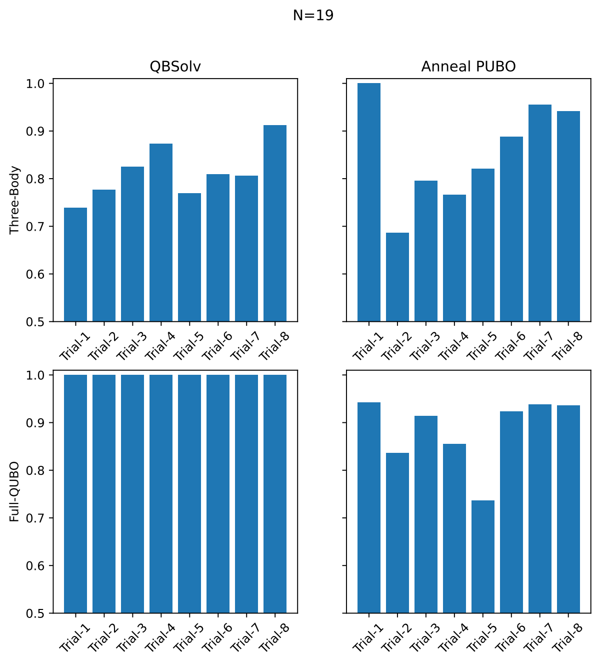


Figure 18: Performance comparison of classical solvers on two types of PCBO problems containing $N = 19$ variables. The y-axes show approximation ratio and the x-axis corresponds to independent evaluation trials. QBSolv finds global optima for all `full-qubo` trials but fails for `three-body-cubo` trials. Anneal PUBO finds the global optimum for one `three-body-cubo` trial.

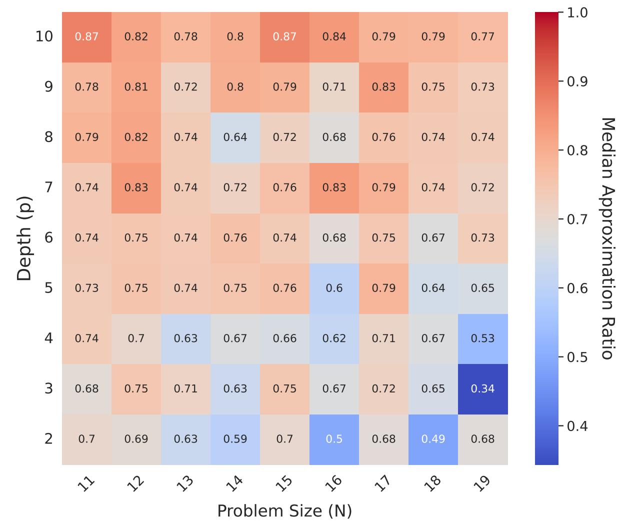


Figure 19: Classical simulations of the RQAOA on the `three-body-cubo` PCBOs of varying size N and depth p. The median approximation ratio is computed over an average of 20 trials for each point, and the RQAOA is evaluated under the strictest solution condition with the cutoff parameter set to 1 [21]. We observe the expected general trend where solution quality increases as the circuit depth p increases [22].

3.3 Simulation Experiments on Perlmutter: Evaluating RQAOA Performance

Our research has made significant progress through a time allocation on the Perlmutter supercomputer, awarded via the QIS@Perlmutter program [35]. This opportunity has allowed us to begin porting our simulation code into a GPU-compatible format, preparing for a future Phase 2. Using Perlmutter, we have collected simulation results (as part of Sub-activities 1.3.1, 1.3.3, and 1.3.5) to analyze how the depth parameter p must scale as a function of problem size to ensure high-quality solutions to PCBO problems — and estimate when the RQAOA solutions become competitive with, and surpass, classical state-of-the-art solvers.

In Figure 18, we evaluate two classical solvers, QBSolv and Anneal PUBO, on two types of PCBO: `full-qubo` (which contains first- and second-order terms) and `three-body-cubo` (which contains first-, second-, and third-order terms). Specifically, the `three-body-cubo` is defined by the expression:

$$\begin{aligned} Q_{\texttt{three-body}}(\boldsymbol{x}, \lambda) = -\lambda \sum_i I(f_i; y) x_i + &\sum_{i,j} \left[I(f_i; f_j; y) - I(f_i; y | f_j) \right] x_i x_j \\ &+ \sum_{i,j,k} \left[I(f_i; f_j; f_k) - I(f_i; f_j; f_k | y) - I(f_i; f_j; y | f_k) \right] x_i x_j x_k \end{aligned} \tag{7}$$

The QBSolv algorithm is an open-source algorithm developed by D-Wave and it solves PCBO problems by strategically dividing a large PCBO into a more manageable set of smaller optimization problems to themselves be solved via the Tabu search algorithm, a powerful algorithm for doing iterative rule-based local search [34]. Anneal PUBO is an implementation of simulated annealing available through the open source

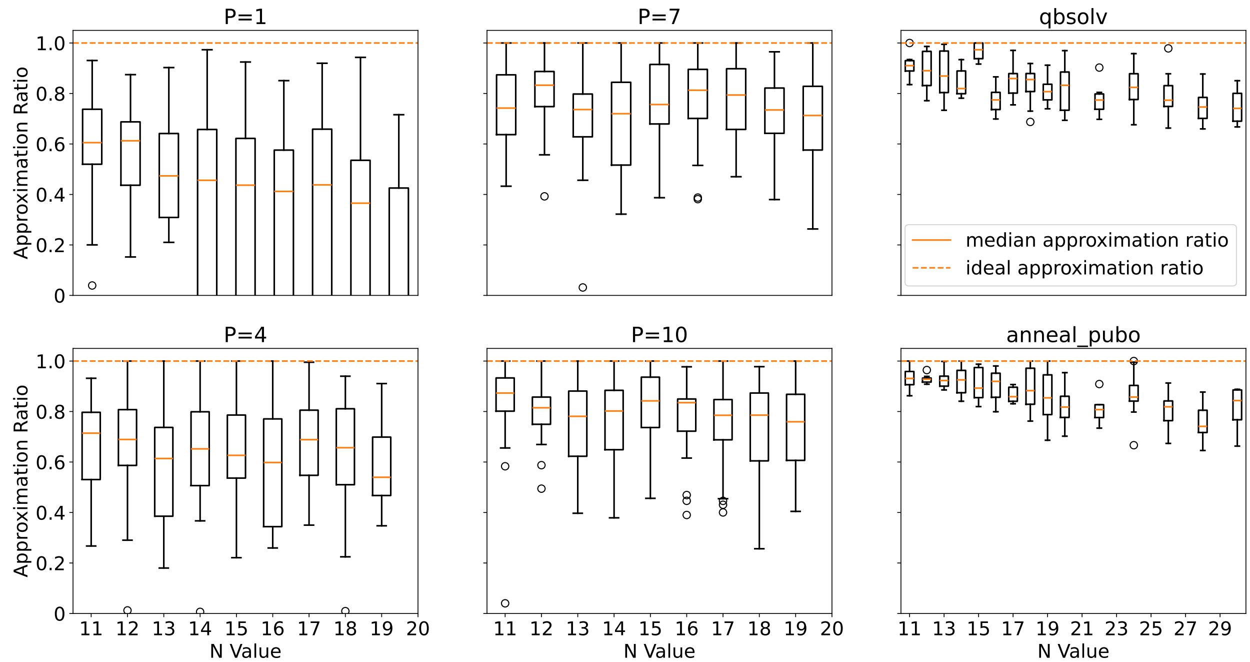


Figure 20: Boxplots displaying the distribution of approximation ratios achieved by RQAOA (for $p = 1, 4, 7, 10$), QBSolv, and Anneal PUBO. Each distribution is formed by an average of 20 independent trials. A range of problem sizes N are shown, up to $N = 19$ for the RQAOA which is the limit of our current single-GPU-node memory and runtime simulation resources, and up to $N = 30$ for the classical solvers. The RQAOA $p = 1$ results contain some negative approximation ratio since these circuits are too shallow to return solutions containing the correct number of features. As the depth p of the quantum circuits increases, the quality of the returned solutions also increases as evidenced by the rising median values. Furthermore, a negative trend in performance can be observed for the two classical solvers as the problem size grows.

`qubovert` Python package [36]. Figure 18 reports the approximation ratio of the solutions returned by the classical algorithms. The approximation ratio is defined as

$$\text{Approx. Ratio} = \frac{Q(\boldsymbol{x}_{alg})}{Q(\boldsymbol{x}_{opt})} \tag{8}$$

where $Q(\boldsymbol{x})$ is the energy assigned to the bitstring returned by the algorithm ($\boldsymbol{x}_{alg}$) and for the global optimum ($\boldsymbol{x}_{opt}$). Examples for various PCBO energy functions are shown in Equations 2, 4, 5, and 6. Notably, the QBSolv algorithm is able to find the global optimum for each `full-qubo` trial problem (bottom row), but is unable to do so for any of the `three-body-cubo` trials (top row). The Anneal PUBO algorithm is able to locate the global optimum for a single `three-body-cubo` trial.

Interestingly, while the classical algorithms seem to easily handle the small `full-qubo` problems, they tend to struggle with the `three-body-cubo` instances, even at relatively small sizes (Figure 18 shows eight $N = 19$ trials and Figure 20 contains evaluations up to $N = 30$ variables). This suggests that the potential for quantum advantage lies in finding higher-quality solutions to the higher-order PCBO problems. Therefore, for the remainder of this section describing our simulations on Perlmutter, we focus on simulating the RQAOA's performance on the `three-body-cubo` PCBOs that contain first-, second-, and third-order terms. The heatmap in Figure 19 provides an overview of these simulations where we plot the median approximation ratio achieved over an average of 20 independent trials (there is some variation in the exact number of trials for each setting of N and p due to differing runtime and memory requirements for the simulations). We observe a clear trend of increasing median approximation ratio with increasing p across all problem sizes N. Furthermore, the RQAOA simulations are performed under the strictest solution condition (cutoff=1) where the algorithm recursively solves each QAOA subproblem all the way down to a single remaining spin

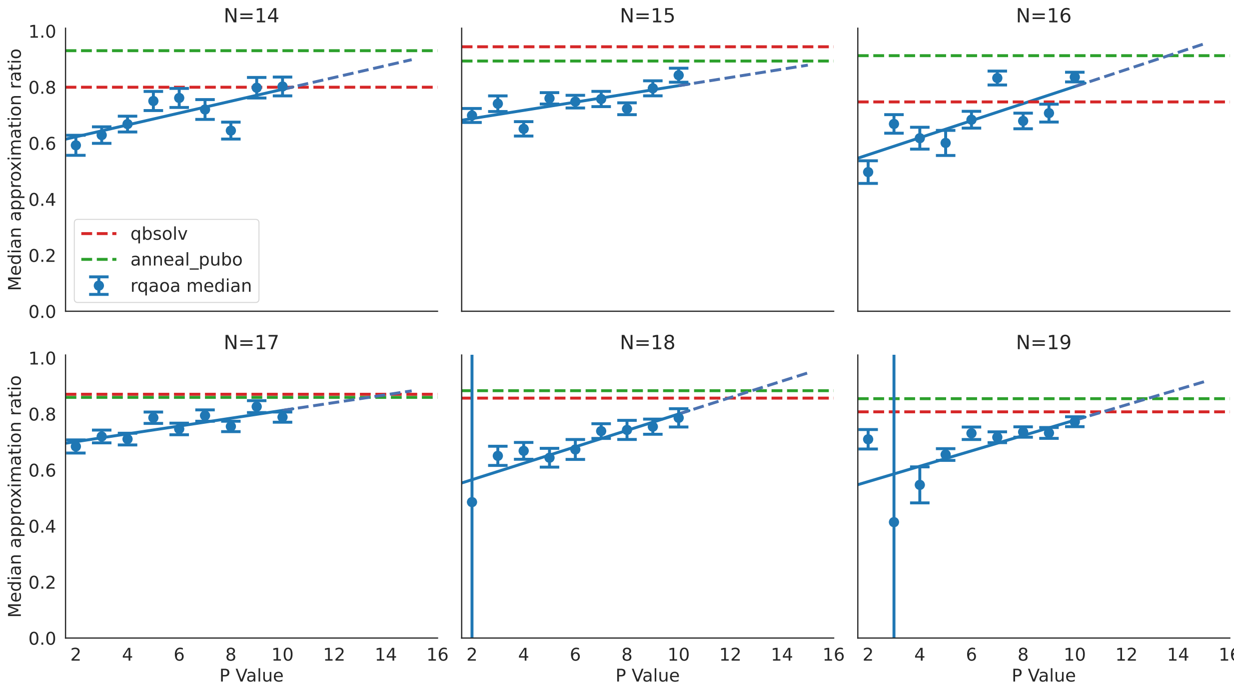


Figure 21: We report the median approximation ratio for RQAOA as a function of increasing p value. Error bars represent the standard error for the estimate of the median. The large error values for the low p values at larger N are a result of the RQAOA returning solutions that violate the feature set size constraint. The larger problems require deeper circuits to return higher quality solutions that abide by the constraint. The dashed blue line extends the linear fit to the RQAOA median values showing the estimated p values required to outperform both of the two classical solvers QBSolv (red) and Anneal PUBO (green). Studying the RQAOA in this higher-depth regime is the focus of our ongoing and proposed Phase 2 work.

variable. This is a stricter setting than the original specification of the RQAOA which suggested that a brute-force classical solver could be used to finish solving the problem once RQAOA had reduced it to a small enough size, and it's likely that this is how RQAOA would be used in practice [21]. For the RQAOA simulation results presented in Figures 19, 20, and 21 we used the COBYLA optimizer (using the default `scikit-learn` settings [24]) initialized with random angles and capped at 1000 optimizer iterations. Note that, the range of possible energy values spans both negative *and* positive values. The ground state (optimal) energy is always negative, but because this is a constrained optimization problem, a positive additive penalty is applied to bitstring solutions with a Hamming weight that differs from the desired number of features, therefore, a negative approximation ratio is possible, but for reasonably large values of p these outliers are less common.

To assess the potential for quantum advantage on the third-order PCBO problems we simulated the RQAOA for a range of circuit depths p and compared its performance against the classical QBSolv and Anneal PUBO solvers on an ensemble of problem instances. We considered different `three-body-cubo` PCBO formulations to select either four or six features for a range of problem sizes up to $N = 19$, which also corresponds to the maximum number of qubits that need to be simulated. Since classical algorithms both return a distribution of solutions, we recorded the best solution returned out of 100 sampled solutions.

After $N = 30$ variables, verifying the approximation ratio via brute force search became challenging, and a trend had become apparent for the classical solvers (see QBSolv and Anneal PUBO in Figure 20). Not only was the quality of the classical solutions dropping, but both solvers were starting to display individual weaknesses. When the QBSolv algorithm is applied to PCBOs containing third-order terms it must add a quadratic overhead of extra variables to convert the problem back into a quadratic formulation, making it untenable for problems with 100-1000 variables. While Anneal PUBO does not incur extra variable overhead, its runtime does increase rapidly, also making it untenable for large numbers of variables. The boxplots in

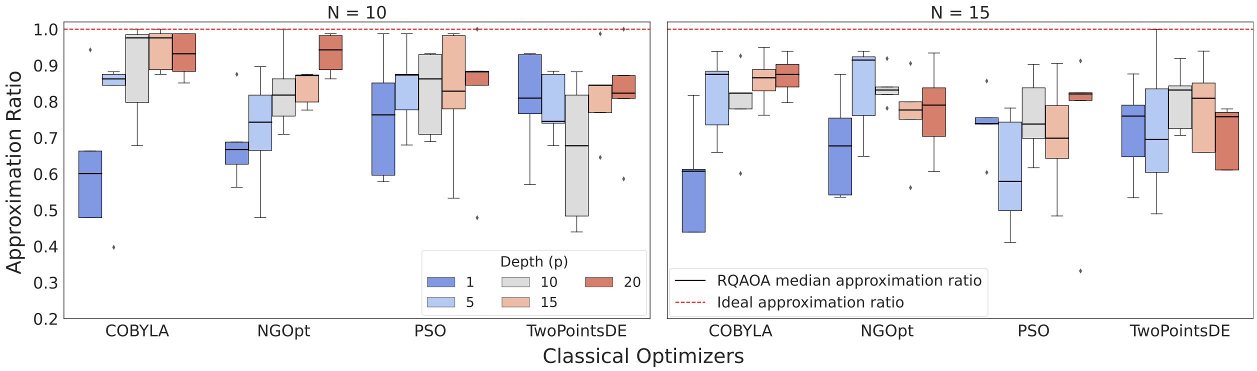


Figure 22: Comparative performance of local (COBYLA) and global (NGOpt, PSO, TwoPointsDE) optimizers for RQAOA on `three-body-cubo` feature selection problems with 10 and 15 features. Results show varying approximation ratios across optimizers and p-values, with potential for improvement through initial parameter tuning and increased computational resources.

Figure 20 do not show an indisputable separation between the quantum and classical solutions; however, they do indicate that for modest values of p, the quantum solver is competitive with the classical ones and that for almost any p, the quantum solver has the potential to find the ground state. Figure 20 also shows that for larger problems, beyond our initial quantum circuit simulations with a limit of $N \leq 19$, there is a clear drop-off in the approximation ratio for the classical solvers, indicating that this problem is indeed difficult in the regime we are targeting.

In Figure 21, we hold the number of variables fixed and show the relationship between p and the median approximation ratio. We observe that as p increases the median approximation ratio increases, sometimes above the classical solution. This investigation allows us to reason that for large problems, RQAOA at modest p depth will outperform classical solvers and stay within the budgeted quantum resources for Phase 3 (see resource scaling plot in Figure 23 for larger qubit and depth projections).

Our Phase 1 simulation experiments indicate that RQAOA shows considerable promise in tackling large feature selection problems, particularly at modest p depths relative to the problem size N. These findings align well with our project's quantum resource constraints and suggest that RQAOA could offer a significant advantage over classical methods in this problem domain. Future work in Phase 2 will involve extending these investigations to even larger problem sizes and deeper quantum circuits (i.e., larger N and p) by leveraging multi-GPU-node simulations. The utilization of HPC resources in our study of the RQAOA is necessitated by the computational intensity of variational experiments, despite the feasibility of simulating *individual* 25- or 30-qubit circuits on standard hardware (e.g., a laptop). RQAOA's iterative nature, requiring repeated execution of QAOA at descending problem sizes, and the variational optimization for *each* of those QAOA subproblems, can quickly lead to $\mathcal{O}(10,000)$ circuits or more that need to be simulated — rendering HPC an essential resource. To ensure scalability in the current and subsequent research phases, we employed the Perlmutter supercomputer, achieving a 64-fold improvement in circuit simulation time through GPU utilization. While implementing HPC presented challenges, it enabled the development of a robust data collection pipeline, laying the groundwork for more extensive quantum simulations in future research. Notably, our current simulations have all been limited to single-GPU-node configurations (on Perlmutter this corresponds to four A100 GPUs), and our immediate near term goals include extending these simulations to *multi*-GPU-node simulations by leveraging the Message Passing Interface (MPI) paradigm. Our approach not only addresses immediate computational demands but also establishes a foundation for advancing quantum algorithm simulation and optimization methodologies.

3.4 Resource Optimization and Estimation

Lastly, as part of Sub-activities 1.3.4 and 1.4.1, we have initiated investigations of improved variational optimizers for the QAOA circuits as well as quantum resource estimations, projecting qubit and circuit depth requirements to align with the Q4Bio target resources. Our investigation and simulations of the RQAOA

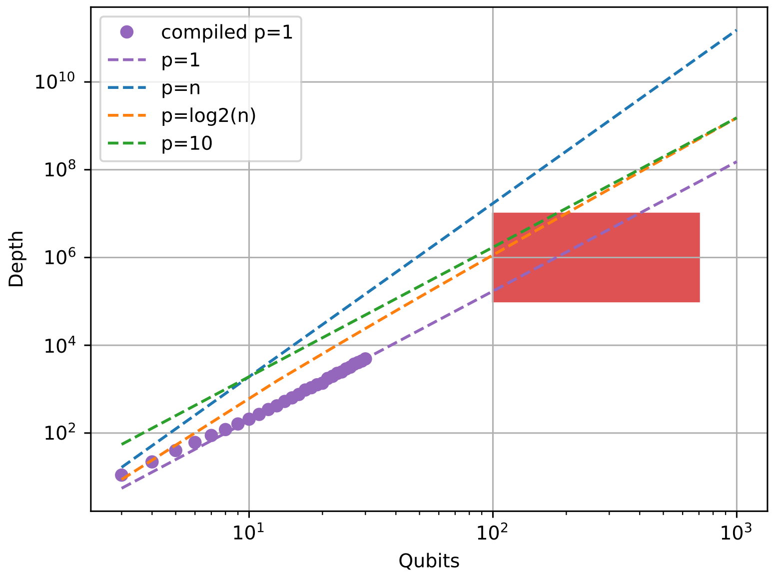


Figure 23: Aligned with Sub-activity 1.4.1 we report the estimated scaling of the quantum circuit depths required for the PCBO-Tournament algorithm. The parameter p controls the number of layers in the QAOA circuits and we plot scaling relations as a function of the problem size n. The Q4Bio target resources are highlighted in the red box, and for all scaling curves, except linear, the quantum resources fit within the target. For linear $p = n$ scaling, we are investigating additional circuit optimizations to lower the depth scaling to fit within the target resources.

applied to the PCBO problems quickly revealed that a primary challenge to scaling this algorithm to larger problem sizes is our choice of the classical optimizer used to tune the variational circuit parameters. The design of more shot-efficient optimizers and other techniques such as warm-starting, parameter concentration, and parameter transferability are widely studied topics in variational quantum algorithms [32, 37, 38, 39, 40, 41, 42, 43], especially for QAOA, and we may therefore leverage this extensive prior work to reduce the total number of calls to the quantum computer and optimize the quantum resources required by the PCBO-Tournament algorithm.

We tested the efficacy of various local and global optimizers for RQAOA with the results shown in Figure 22. While COBYLA served as the primary local optimizer, we also explored global optimizers from the Nevergrad suite, including NGOpt, Particle Swarm Optimization (PSO), and TwoPointsDE. The experimental design incorporated five independent trials per p value and optimizer, with randomized initial angles for each QAOA iteration and a consistent cap of 1000 optimizer iterations. Notably, we observed that these optimizers could achieve high approximation ratios, albeit with some variance. This variability suggests potential benefits from more thoughtful initial parameter selection, a avenue we intend to explore in Phase 2 of our research. Furthermore, the utilization of additional GPU resources is expected to enhance optimizer performance through fine-tuning of meta-parameters and experimentation with higher iteration limits, moving beyond the current out-of-the-box configurations. These findings not only elucidate the current performance of RQAOA under various optimization schemes but also highlight promising directions for future research to improve the algorithm's efficiency and reliability in solving complex feature selection problems.

Figure 23 shows the circuit depth for various QAOA circuits, assuming all-to-all connectivity, with p scaling as different functions of problem size n. Note that, in the case of PCBO feature selection, the number of features n is also equal to the number of qubits required. The circuit depth estimations in Figure 23 represent upper bounds on the true circuit depth. The depth is estimated by constructing the QAOA ansatz for a single $p = 1$ layer using the greedy parity synthesis method developed by one of our team members [44], and then multiplying the depth of this circuit by the desired value of p to produce the final estimation. We also include "compiled $p = 1$" data points representing cases where we constructed the whole circuit and computed its depth to ensure the accuracy of our estimations. Our estimation method does not account for additional gate cancellation opportunities between gates in different layers of the QAOA circuit. Our analysis indicates that sub-linear scaling of p with n will place us within the target resources. Importantly, we are also exploring additional optimizations, such as coefficient thresholding, to further reduce gate count and lower circuit depth to meet target resources.

4 Key Innovations and Artifacts

1. An initial draft of our preprint, "On the potential for quantum advantage in feature selection applications for multimodal cancer data", has been made available to the reviewers through our shared Google Drive with Wellcome Leap [10]. This paper contains the results of our algorithm development performed for Milestones 1 and 2 of Phase 1, and will be updated to reflect ongoing evaluations to be included in the final Phase 1 report. Highlights include:

   - We have developed scalable, classical methods for pre-processing our multiple modes of data in a way that allows us to control the computational resources required in the downstream feature selection and learning steps. This includes methods for discretizing continuous mRNA features to allow for more efficient data loading on a quantum computer, as well as classical deep learning methods to pre-process the gigapixel-sized pathomics whole-slide images.
   - Our methodology is novel in the way it makes use of classical-quantum resources to select feature sets within the framework of constrained combinatorial optimization problems.
   - Our empirical results (see Figure 10) show that our hybrid algorithms are capable of selecting smaller feature sets which outperform the classical algorithms (e.g., random-forest-based feature importance [23], mutual information-based univariate feature selection, mifs-nd [25]) used as baseline comparisons.
   - A key innovation in our algorithm design was the divide-and-conquer approach which enables feature selection on data sets with an arbitrary number of features using a fixed amount of quantum resources.
   - Additionally, the inclusion of higher-order correlations between features to select relevant feature sets is a novel innovation for quantum optimization applications. This technique can be scaled to any order of correlations, as a way to explicitly capture the underlying biology, depending on available resources. Further, we have demonstrated the ability of PCBOs with third-order interaction information to select better predictors of predicted complete response in the ISPY data than PCBOs with only first- and second-order mutual information. This suggests the need for higher-order information-theoretic tools for feature selection in computational oncology, tools which are are computationally expensive for classical computers to implement. We are currently developing recursive quantum algorithms for feature selection, and our evaluations are some of the first experiments to test whether such problems are amenable to quantum advantage.

2. We are currently developing techniques to measure the suitability of classical data sets for quantum advantage. Already, we have a preliminary proof placing bounds on classical machine learning performance for data sets which satisfy certain conditions that make them provably hard for such classical algorithms. Our project is focused on building specific implementations for cancer data sets which would prove the existence of advantaged quantum models empirically.

3. Our published manuscript "Quantum Computing for Oncology", highlighting and acknowledging our Q4Bio supported innovation, is the first quantum computing-focused publication to appear in *Nature Cancer* [45]. It introduces general quantum computing concepts to the broader oncology community and highlights the potential application areas where we expect quantum computation to have the greatest impact, emphasizing this project focus as one area of significant promise.

4. Our team wrote an article "Co-Design of Hybrid Quantum-Classical Applications in Multimodal Cancer Research" which appeared in the ACM SIGARCH Computer Architecture Today blog [46]. The article emphasized the role that interdisciplinary collaboration can have in the search for applications with quantum advantage, and we highlighted our Q4Bio project as one such example.

Ultimately, the key innovation in our work is identifying feature selection for multimodal cancer data as a promising area for near-term quantum advantage. By formulating this as a PCBO problem that includes higher-order feature correlations, we've uncovered a domain where classical algorithms quickly become intractable. Our preliminary results show that even for modest problem sizes of limited numbers of variables, classical heuristics struggle to solve PCBOs with third-order terms. This suggests that this is an area of significant need in current biological machine learning paradigms, making it a particularly exciting area for quantum-classical hybrid algorithms. If quantum approaches can efficiently solve these complex PCBOs, it could lead to significant improvements in biomarker identification and cancer diagnostics, showcasing a concrete example of quantum computing's potential impact in oncology.

5 Addressing Phase 1 Evaluation Criteria

5.1 Potential Impact in Bio

1. Efficient Identification of Informative, Interpretable Biomarkers: Selecting small feature sets that are highly informative of clinically relevant attributes would have important clinical impacts, including reducing biomarker cost and improving clinical interpretability.

2. Complex Biological Insight: Capturing higher-order correlations and multimodal data interactions will potentially uncover novel biological relationships, leading to new modeling and treatment strategies.

3. Improved Patient Stratification and Personalized Medicine: Enhancing accuracy in cancer subtyping and treatment prediction will lead to more effective targeted therapies.

4. Data Scarcity Solution: Enriching model quality in data-limited scenarios that are common in cancer research will reduce overfitting and overparameterization issues seen in many prior works.

5. Quantum-enabled Feature Selection as a Learning Strategy for Biological Systems: Quantum-classical hybrid algorithms enables feature selection to potentially become a powerful learning strategy, with potential applications in diverse biological systems.

6. Multimodal Cancer Data Analysis as a Quantum Computing Target: Our demonstration of effective quantum-classical strategies in multimodal cancer data analysis highlights this biomedical research area as a target for future quantum algorithms development, with potential major long-term impacts.

7. Bridging the Research-Clinic Gap: By making complex multimodal data analysis relevant for clinical applications, this work may potentially speed up the adoption of precision oncology approaches.

5.2 Improvement in Computational Resources

Our research has made significant strides in addressing the computational resources required for quantum algorithms applied to feature selection on multimodal data sets. The number of qubits required by the PCBO-Tournament algorithm scales linearly with the number of features considered in each PCBO *subproblem*, utilizing one qubit per feature. Thanks to the divide-and-conquer approach, breaking the full input feature set into smaller subproblems, the number of qubits does not scale with the total number of features in the data set. This enables a quantum processor with a fixed number of qubits to select features from arbitrarily large data sets, overcoming a key limitation in quantum computing applications. Additionally, our compilation results in Figure 12 indicate that there is much potential for gate cancellation and exploitation of parallelism in the QAOA circuits used to solve the PCBO subproblems, especially when taking into account hardware-specific information such as qubit connectivity and native gate set [26].

5.3 Advances in Phase 1

1. A novel method for pre-processing mRNA features; we present a quantile-based method for discretizing continuous mRNA expression values. In principle, this method is robust to the splitting of data into training and testing subsets (i.e., the quantile computation is robust to data subsetting when the data are unimodal). It is easy to compute, and it enables meaningful feature selection, as demonstrated by our preliminary experiments.

2. Development and evaluation of novel methods for summarizing and discretizing pathomics data for quantum processing. We have developed custom methods to classically pre-process and condense pathomics information into data formats more conducive for quantum processors. We have compared our custom methods against recently published literature and developed synergistic approaches that leverage both internally and externally developed techniques.

3. We have developed a novel hybrid algorithm for selecting features in multimodal cancer data sets. Notably, the time complexity of our hybrid algorithm is comparable to classical approaches (see Table 3). However, in feature selection tasks, solution quality often takes precedence over time-to-solution. This is because feature selection is typically a one-time process for a given data set, making it advantageous to invest more time if it yields higher quality results. In this regard, our preliminary findings suggest that the PCBO-Tournament algorithm has the potential to select smaller, higher quality feature sets (see Figures 10 and 16). We anticipate further improvements over classical approaches as we incorporate higher-order terms and the quality and scale of quantum devices advance (allowing for larger and deeper circuit execution – see the trendlines in Figure 21). It is important to note that this comparison essentially evaluates different heuristics. As such, the most definitive approach is a direct, head-to-head comparison. We are actively working towards this goal, with plans for HPC simulation in the near term and hardware evaluation in the future.

5.4 Enhanced Understanding

Our research has significantly enhanced understanding of how quantum-enhanced feature selection can impact the analysis and use of complex, multimodal cancer data. Our methods to sub-select informationally-rich and non-redundant feature sets help address overparameterization and overfitting issues common in classical approaches, ultimately enabling more generalizable models across diverse patient cohorts. Crucially, our results allow the evaluation of novel gene sets and combined genomic-expression-pathology features that may have been previously overlooked by traditional methods. This capability opens new avenues for exploring how these feature combinations relate to underlying disease mechanisms. Identifying informative feature sets not only improves predictive power but also generates hypotheses about biological interactions that can be further probed through targeted experiments. For instance, identification of unexpected correlations between genetic markers and pathological features could guide wet-lab studies to uncover new disease pathways or biomarkers. This iterative process between computational prediction and biological validation has the potential to accelerate discoveries in cancer biology and drive more precise diagnostic and therapeutic strategies. Ultimately, our approach bridges theoretical quantum computing, systems biology approaches, and practical oncology applications, laying the groundwork for a new paradigm in data-driven cancer research.

Our efforts developing and optimizing quantum algorithms for feature selection on multimodal cancer data will also advance broader understanding of the utility of variational quantum algorithms. While previous studies have been limited to small qubit numbers and shallow circuit depths, our work pushes the boundaries by evaluating the performance of quantum algorithms, such as RQAOA, at larger and deeper circuit sizes. We are leveraging high-performance computing resources to explore the large qubit and deep circuit depth regimes, areas where theoretical performance bounds for heuristic algorithms have been notoriously difficult to establish. This approach not only aims to determine the practical utility of these algorithms, but also develops novel techniques to enhance their performance and address known challenges, such as trainability, which may be of broader interest for other applications. Our research encompasses advanced circuit optimization methods and strategies to improve trainability, such as warm starting and custom optimizers tailored for variational circuits. By doing so, we bridge the gap between theoretical potential and practical implementation, paving the way for more effective quantum solvers for a broader range of optimization applications.

5.5 Feasibility for Next Phases

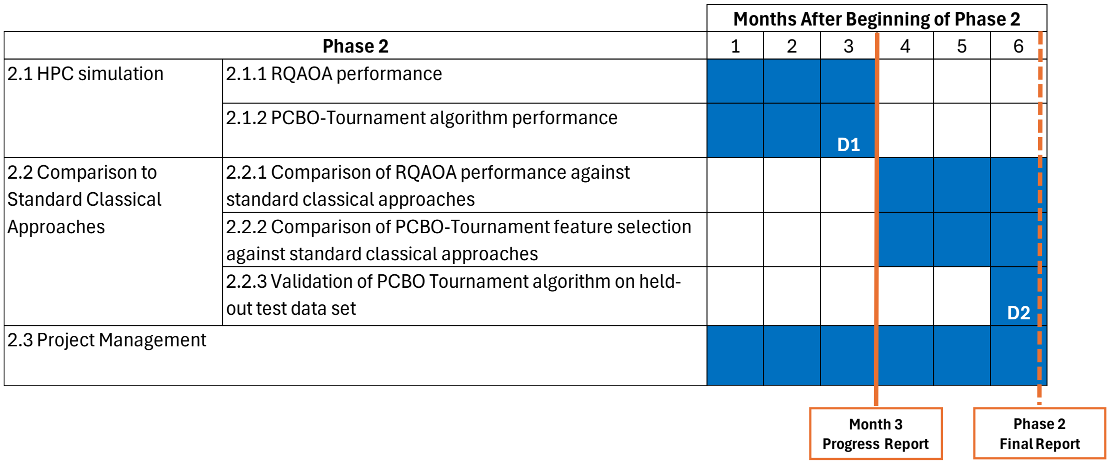


Figure 24: Current draft of Phase 2 GANTT chart.

To demonstrate feasibility of success in the next two phases of the Q4Bio program, we are pursuing several key strategies across simulation, algorithm optimization, model validation, hardware selection, and hardware-specific compilation. A draft of our planned Phase 2 activities is shown in Figure 24. We are already in the midst of developing, evaluating, and expanding simulations of the RQAOA algorithms on the NERSC Perlmutter supercomputer through their recent QIS@Perlmutter program [35]. Access to these GPU resources has been crucial for accelerating the design of our feature selection algorithm. It has also allowed us to work through the various bugs and challenges that have arisen while porting our original code into a CUDA-Q framework that leverages GPU resources. In practice, these issues can sometimes take multiple weeks to resolve so addressing them now will allow us to make the most efficient use of the Phase 2 period and we will be able to quickly scale up our simulations to include *multiple* GPU nodes on the Perlmutter supercomputer — **allowing us to study the performance of the RQAOA in larger qubit and circuit depth regimes beyond the current state-of-the-art**.

To this end, we plan to request a larger Perlmutter allocation to support this expansion, develop multi-node simulations enabled via the Message Passing Interface (MPI), and ensure continued access to HPC resources throughout the duration of Phase 2. For reference, our current QIS@Perlmutter allocation runs from March 21, 2024 through January 14, 2025. Furthermore, we have worked closely with Mark III and Nvidia to determine the specs of the HPC systems that we intend to purchase with the HPC funding made available as part of Phase 2. Our simulation approach will leverage several strategies, such as QAOA parameter setting [32] and gate fusion techniques which build on prior work by NVIDIA [47], in order to further increase efficiency and allow us to scale up to larger numbers of qubits and deeper quantum circuits.

As shown in Figure 3, our team is actively preparing and collecting additional multimodal cancer data sets that will enable us to effectively validate the performance of our feature selection algorithms. Curating such high quality, multimodal cancer data sets is important not only for validating the performance of our algorithms, but is also of broader interest to the cancer research community where they may aid in the development of novel multimodal diagnostic and predictive techniques. Testing our models on these new data sets is an important step towards verifying performance of our hybrid methods on more realistic, clinically-relevant applications such as predicting tumor response to treatment, bringing us closer to realizing quantum advantage for practical and impactful human health applications.

We are also currently considering the steps necessary to ensure successful hardware evaluations in a future Phase 3. In terms of hardware selection, we are pursuing a multi-pronged strategy. We have the capability to employ D-Wave quantum annealers to solve PCBO feature selection problems at a scale comparable to current state-of-the-art classical solvers (i.e., problems containing $\mathcal{O}(1000)$ variables). We have reached out to D-Wave and received a positive response affirming their interest in support such experiments. Furthermore, we plan to target Quantinuum's latest devices through the ORNL QCUP program [48] and leverage our direct relationship with the company. Both hardware platforms offer unique advantages: D-Wave's annealers support a large number of qubits enabling us to study larger problems comparable to classical state-of-the-art solvers, whereas Quantinuum's trapped ion quantum computers possess all-to-all connectivity and superior gate quality allowing us to execute deeper quantum circuits. This diverse hardware approach will allow us to evaluate and optimize our algorithms across multiple quantum computing platforms, enhancing the robustness and applicability of our results. Additionally, Infleqtion's Superstaq compiler will play a key role in optimizing the compilation of our RQAOA circuits for different hardware backends. We plan to study how compilation to different hardware backends affects overall circuit depth, with a particular focus on extracting maximum parallelism. This research is well aligned with the expertise of the Infleqtion quantum software team which has extensive experience developing hardware-specific and pulse-level optimization schemes to maximize the performance of near-term quantum hardware [26] .

6 The Potential Impact of Our Work

Our work on quantum biomarker algorithms for multimodal cancer data addresses critical limitations in current classical approaches. Our *Nature Cancer* paper and SIGARCH blog post demonstrate the utility of quantum codesign in oncology research, highlighting a new set of problems that are important but less well understood outside the biomedical sphere. By bringing these challenges to the attention of broader computational fields, we are fostering interdisciplinary collaboration and innovation.

Traditional multimodal learning methods often struggle with the high dimensionality and heterogeneity of cancer data sets, particularly when integrating genomic, transcriptomic, and pathomics data. Classical algorithms frequently rely on simplistic linear models or weak heuristics that fail to capture complex, non-linear relationships across data modalities. Given the sample limitations inherent to biomedical research and the exponential growth in dimensionality introduced by evaluating multiple data modalities simultaneously, many studies publish findings that can be partially attributed to overfitting to specific data sets [33]. Our hybrid quantum-classical approach, leveraging quantum resources for combinatorial optimization in feature selection, has the potential to more effectively explore the vast search space and identify informative, non-redundant feature sets that are predictive of cancer outcomes, thereby increasing the ability to develop meaningful and generalizable models.

A key innovation in our work is the ability to incorporate higher-order correlations between features across different data modalities. Classical methods often struggle with this task due to computational limitations and the curse of dimensionality. The PCBO-Tournament approach can efficiently handle these higher-order interactions, potentially uncovering subtle but significant biological relationships that may be missed by traditional methods. This could lead to the identification of novel multimodal biomarkers that more accurately reflect the complex underlying biology of cancer, improving diagnostic accuracy and treatment stratification.

Furthermore, our research addresses the critical challenge of limited training data in cancer research. Classical machine learning models often require large amounts of data to generalize well, which is not always available in oncology studies. Our quantum-enhanced approach aims to make more efficient use of limited data by identifying the most informative features across modalities. This could be particularly impactful in rare cancer types or when studying specific molecular subtypes where data scarcity is a significant hurdle. By potentially reducing the number of features needed for accurate predictions, our methods could also lower the barriers to clinical implementation of multimodal biomarkers, making personalized medicine more accessible and cost-effective.

Our project highlights the benefit of interdisciplinary teamwork, bringing together experts from quantum computing, oncology, and data science to tackle complex biomedical challenges. This collaborative approach not only enhances the quality of our research but also serves as a model for future interdisciplinary efforts in quantum computing and healthcare.

6.1 Extensions and Trainee Career Development

During Phase 1, our team described the vast range of applications of quantum computing to oncology research in a published manuscript in *Nature Cancer* [45]. Furthermore, our team has together prepared additional, unique, non-redundant, self-sustaining grant applications for NIH and ARPA-H. We are committed to leveraging the high-impact work described here with our clinical team members in order to achieve real application and eventual transformative change in cancer research. Our research group has integrated personnel at multiple career stages. Multiple PhD graduate students have been directly involved in the project. Furthermore, former medical student Dr. Sid Ramesh has performed substantial research on this grant and was offered a position in the prestigious University of Chicago Physician Scientist Training Program to continue quantum computing research with medical applications.

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